The Heat and Transdisciplinary (HE²AT Center)



Health African Center

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Individual Participant Data meta-analysis to quantify the impact of high ambient temperatures on maternal and child health in Africa

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Acronyms and definitions

Acronym	Definition
AIDS	Acquired Immune Deficiency Syndrome
AOD	Aerosolized Optical Depth
BP	Blood pressure
CDC	Centers for Disease Control
CHANCE	Climate-Health Africa Network for Collaboration and Engagement
Co-PI	Co-Principal Investigators
CSAG	Climate System Analysis Group
DMAC	Data Management and Analysis Core
DMP	Data Management Plan
DoH	Department of Health
DRS	Data Reference Syntax
DSA	Data Sharing Agreements
DS-I Africa	Harnessing Data Science for Health Discovery and Innovation in Africa
ECMWF	European Centre for Medium range Weather Forecasts
EDCTP	European and Developing Countries Clinical Trials Partnership
FIC	Fogarty International Center
GAM	Generalized Additive Model
GANs	General Adversarial Networks
GCRO	Gauteng-City Region Observatory
HE^2AT	Heat and Health African Transdisciplinary
HIV	Human Immunodeficiency Virus
HPC	High Performance Computing
ICH	International Conference on Harmonisation
IBM-PAIRS	IBM Physical Analytics Integrated Data and Repository Services
IPCC	Intergovernmental Panel on Climate Change
IPD	Individual Participant Data
MTCT	Mother to child transmission
NIH	National Institute of Health
NDVI	Normalized Difference Vegetation Index
PI	Principal Investigator
PM	Particulate Matter
POPIA	Protection of Personal Information Act
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta Analysis
PROM	Premature Rupture of Membranes
RHI	Reproductive Health and HIV Institute
RoB	Risk of bias
RP1	Research Project 1
RP2	Research Project 2
RPs	Research Projects
SC	Steering Committee
TEC	Training and Engagement Core
UCT	University of Cape Town
WHC	Wits Health Consortium
WHO	World Health Organization
WWARN	Worldwide Antimalarial Resistance Network

Term	Definition
Newborn	0 to 72 hours
Neonate	0 to 28 days of age
Infant	0-1 years old
Postpartum	0-42 days after childbirth (in mother)
Intrapartum	During labour and delivery/childbirth
Cohort study	An observational study in which a defined group of people (the cohort) is followed over time. The outcomes of people in subsets of this cohort are compared, to examine people who were exposed or not exposed (or exposed at different levels) to a particular intervention or other factor of interest (1)
Clinical trial	A research study in which one or more human subjects are prospectively assigned to one or more interventions (which may include placebo or other control) to evaluate the effects of those interventions on health-related biomedical or behavioural outcomes (2)

Study synopsis

Rationale: Globally, recognition is growing of the harmful impacts of extreme heat on health, across a range of population groups, including pregnant women and children. There remain, however, major gaps in evidence on the extent to which high temperatures increase the risks for adverse health outcomes and which outcomes are most affected. Gaps in evidence are especially large in Africa. An individual participant data (IPD) meta-analysis entails systematically locating, appraising, transforming, and analysing individual data from multiple studies which have a common outcome of interest. Therefore, unlike classical systematic reviews, the IPD uses participant-level data, rather than simply using aggregate-level data extracted from a publication. This approach can overcome many of the biases of classic systematic reviews of published data, and the challenges in understanding heterogeneity in findings, and methodological diversity across published studies. Additionally, analysing pooled participant-level data from multiple settings and time periods can mitigate many of the limitations of analysis that include individual databases from a single location and time, most especially in limitations of study power and generalisability. A detailed understanding and quantification of the linkages between heat and health is essential for developing solutions to this critical policy and research area.

The study forms parts of DS-I Africa which aims to make optimum use of existing data resources across Africa to address the most pressing health concerns on the continent. The study constitutes one of two research projects within the HEat and HEalth African Transdisciplinary Center (HE^2AT Center) project funded through the DS-I Africa Program.

Objectives: The primary objective is to use innovative data science approaches to quantify the current and future impacts of heat exposure on maternal and child health in sub-Saharan Africa.

The secondary objectives are:

1. To acquire, collate and integrate prospectively collected data from cohort studies and randomized trials on maternal and child health in sub-Saharan Africa
2. To develop a collaboration between study groups and the IPD investigators
3. To link health outcome data spatially and temporally with weather and other environmental data, as well as socio-economic and other data
4. To utilize classic statistical and novel machine learning approaches to understand and quantify the impact of heat exposure on pregnant women and children
5. To document in detail the variations in the relationship between heat exposure, and maternal and children health outcomes across different settings, climate zones and population sub-groups.

Study Design: Systematic review and meta-analyses of individual participant data.

Methods: We will systematically identify eligible studies through a mapping review, searching through data repositories, as well as suggestions from experts and investigators in this field. Inclusion is based on study- and individual-level criteria. To be eligible, studies need to have collected longitudinal data, have enrolled at least 1000 pregnant women from sub-Saharan Africa in the period January 2012 to June 2022, have collected data on key outcomes of the review, and have relevant ethics approvals in place. At an individual level, participants need to have been recruited during pregnancy or intrapartum and have data available on date of birth and location. We will obtain data from open sources, data repositories or through a collaboration with investigators or

data holders from eligible studies. The individual datasets will be harmonised through the recoding of raw individual participant data into a common set of variables. A one and/or two-stage analysis method will be adopted whereby, in the first-stage, each study is analysed individually. Then, in the second stage, the data from the individual studies are aggregated to provide a pooled estimate of effect. If heterogeneity between studies is high, then aggregation across studies may not be done, or only be done in particular groups of studies that share common characteristics. Various traditional models such as time-series analysis, time-to event analysis and generalised additive models as well as novel machine learning approaches will be used to quantify associations between exposure to high ambient temperatures, and adverse maternal and child outcomes.

Ethical and legal considerations: There is minimal risk to study participants. Participant privacy is protected as far as possible through the anonymisation of data for analysis, and security measures such as limited access to data, data encryption upon transfer and data storage in secure, password-protected servers. Informed consent procedures for the original studies will be assessed, and reconsenting or informed consent waivers considered, if required.

Dissemination: Findings will be disseminated through conferences and journal publications, as well as related policy and research fora.

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PROSPERO registration: The systematic review to locate eligible studies entitled: 'Systematic mapping of longitudinal studies involving pregnant women in sub-Saharan Africa: preliminary data for a data science project on climate change and maternal health using Individual Participant Data' has been registered (PROSPERO CRD42020214637).

The protocol has been registered on PROSPERO CRD42022346068

1. Study partners and investigators

1.1 Study partners

Name of institution	Country	Address
Centre for Sexual Health and HIV & AIDS Research (CeSSHAR)	Zimbabwe	4 Bath Road, Belgravia, Harare, Zimbabwe
Climate System Analysis Group, University of Cape Town	South Africa	Dept. Of Environmental and Geographical Science, University of Cape Town, Rondebosch, Cape Town, South Africa, 7700
IBM Research Africa	South Africa	Wits Research Lab, 47 Juta Street, Braamfontein, Johannesburg, South Africa, 2001
University of Michigan	United States of America	UM Center for Global Health Equity, North Campus Research Complex. 2800 Plymouth Road Building 100. Ann Arbor MI 48105 USA
University of Washington	United States of America	Center for Health and Global Environment (CHanGE), Department of Environmental and Occupational Health Sciences, School of Public Health, University of Washington, Box 354695, Seattle, Washington, USA
University Peleforo Gon Coulibaly	Cote d'Ivoire	UPGC BP 1328 Korhogo, Côte d'Ivoire / CSRS,01 BP 1303 Abidjan 01, Adiopodoumé- Km 17-Route de Dabou, Côte d'Ivoire
Wits RHI, University of the Witwatersrand	South Africa	Hillbrow Health Precinct, 22 Esselen Street, Hillbrow, Johannesburg, South Africa, 2001

1.2 Study investigators

Principal Investigators

Name	Institution	Contact details
Stanley Luchters	CeSHHAR	stanley.luchters@ceshhar.co.zw stanley.luchters@lstmed.ac.uk
Matthew Chersich	Wits RHI	Matthew.Chersich@witsphr.org

Co-Principal Investigators

Name	Institution	Contact details
University of Cape Town		cjack@csag.uct.ac.za
Darshnika Pemi Lakhoo	Wits RHI	darshnika.lakhoo@witsphr.org
Sibusisiwe Makhanya	IBM Research Africa	sibusisiwe.makhanya@ibm.com

Co-Investigators

Name	Institution	Contact details
Abdoulaye Tall	University Peleforo Gon Coulibaly	abdoulaye.tall@csrs.ci
Adja Ferdinand Vanga	University Peleforo Gon Coulibaly	ferdinand.vanga@upgc.edu.ci afvanga@yahoo.fr

Akbar Waljee	University of Michigan	awaljee@umich.edu
Bonnie Joubert	National Institute of Health	bonnie.joubert@nih.gov
Brama Koné	University Peleforo Gon Coulibaly	brama.kone@gmail.com
Christopher Boyer University of Washington		cboyer10@uw.edu
Cherlynn Dumbura	CeSHHAR	cherlynn.dumbura@ceshhar.co.zw
Craig Parker	Wits RHI	Craig.parker@witsphr.org
Euphemia Sibanda	CeSHHAR	euphemia@ceshhar.co.zw
Gciniwe Dlamini Baloyi	IBM Research Africa	gciniwe.dlamini@za.ibm.com
Gloria Maimela	Wits PHR	Gloria>maimela@witsphr.org
Guéladio Cissé	University Peleforo Gon Coulibaly	gueladio.cisse@upgc.edu.ci gueladio.cisse@swisstph.ch
Ijeoma Solarin	University of Witwatersrand	ijeoma.Solarin@witsphr.org
Ji Zhu	University of Michigan	jizhu@umich.edu
Kimberley McAllister	National Institute of Health	mcallis2@niehs.nih.gov
Lisa van Aardenne	University of Cape Town	lisa@csag.uct.ac.za
Maliha Ilias	National Institute of Health	maliha.ilias@nih.gov
Pierre Kloppers	University of Cape Town	pkloppers@csag.uct.ac.za
Piotr Wolski	University of Cape Town	wolski@csag.uct.ac.za
Relebohile Motana	Wits RHI	rmotana@wrhi.ac.za
Webster Mavhu	CeSHHAR	webster@ceshhar.co.zw

2. Background and rationale

Notwithstanding the dangers of novel infectious agents, such as COVID-19, climate change remains the biggest global health threat of the 21st century (3, 4) and the greatest challenge facing this generation (5). While changes in climate manifest in multiple ways, each of these are underscored by the steady rise in global temperatures, principally brought about by the release of greenhouse gases from human activities. In the past 100 years the mean temperature of the world has risen by more than 1°C and by as much as 2°C in many parts of Africa (4, 6, 7). Even if carbon emissions were to decline markedly, further temperature escalation is considered inevitable. The IPCC 6th Assessment report in 2022 concluded that there is a high likelihood that temperature increase will exceed 1.5C by 2050 (8). Further, an estimated 54% of the global population will be exposed to more than 20 days of deadly heat per year by 2100 (9). While the decline in human activities during the COVID-19 pandemic reduced CO_2 emissions globally by around 9%,(10) overall CO_2 levels in the atmosphere continued their inexorable rise, reaching levels of 419 parts per million in March 2022, 100 parts per million above CO_2 levels in 1960 (11). Averting further temperature increases will thus require a societal shift considerably larger than seen during the global slowdown that accompanied the COVID-19 pandemic. Such a shift will need to be underscored by innovative data-driven technologies and solutions, tapping into existing data reserves and novel analytical approaches.

2.1 Heat exposure impacts on health

Heat exposure has a wide range of deleterious effects on health, both direct and indirectly. The indirect impacts of rises in temperature on the burden of infectious diseases are well documented, such as an expansion in geographical range of malaria or dengue fever (12-14). Indirect impacts extend beyond infections, however, and include, for example, increased soil drying and evaporation resulting in crop failure, livestock deaths, food insecurity and malnutrition. Heat also affects health indirectly through fomenting wildfires (15). These fires leave marks that are often long-term, encompassing loss of animal and plant species, direct fire injuries, destruction of infrastructure, malnutrition and poverty (16).

Direct impacts on health of exposure to heat waves and other forms of extreme heat have not entered scientific and public discourse as much as these indirect impacts, though are of no less significance. Many of the temperatures presently being recorded in Africa and those projected in the next decade are already close to the limits of human survival, or 'livability' (7, 9). Temperatures in many areas already exceed levels where physical labour or 'workability' is not possible for parts of the year, with major consequences for the health of workers and workplace performance (17). High ambient temperatures during summer months and discrete heat extremes (such as heatwaves) result in considerable excess mortality and morbidity (18). It is important to note, however, that heat exhaustion and stroke, the classic manifestations of heat exposure are relatively rare. More common manifestations of heat exposure include increased mortality rates among the elderly or those with chronic non-communicable diseases, (19) adverse pregnancy and birth outcomes such as preterm birth, low birth weight and stillbirths, (20) aggression, violence and suicide (21). Rates of visits to emergency rooms and hospitalizations increase during heatwaves, with concomitant escalation in healthcare costs (22). The mental health sequelae of acute or more prolonged periods of heat exposure are considerable, including generalized anxiety, depression and what is commonly referred to as eco-anxiety (18, 21). Patterns of temperature and health outcomes are mostly U-

shaped, with the base of the curves commonly at relatively mild temperatures around 20°C in most tropical and sub-tropical areas, with curves becoming progressively or exponentially steeper with each increase in temperature degree.

2.2 Heat exposure impacts on pregnant women and infants

High ambient temperatures may overwhelm the capacity of maternal thermoregulatory mechanisms to dissipate heat in pregnancy and labour. The physiological and anatomical changes in pregnancy that challenge maternal thermoregulation peak in advanced pregnancy and labour. These include the considerable heat generated by foetal metabolism, strain from additional weight gain in pregnancy, fat deposits that retain heat, and the major exertion of labour and childbirth which generates significant amounts of heat. The increased thermal load makes it difficult for pregnant women or women in labour to maintain a normal temperature range during periods of high ambient temperatures. We highlight four interconnected pathophysiological processes which may explain the harmful impacts of heat.

The first pathway involves elevated maternal body temperatures when thermoregulation capacity is exceeded during pregnancy or labour, or women have a fever. In normal labour, maternal temperature increases by about 0.02°C per hour, but at a considerably faster pace among women who are delivering their first child (23), have prolonged labour (24)[] or who are obese (25). Additionally, maternal core temperature increase during fever from infections, following epidural analgesia (26), or if mechanisms such as sweating and fast breathing cannot compensate for the additional burden of internally generated heat during periods with high ambient temperatures. Because foetal temperature is consistently 0.3-0.5°C higher than the maternal core temperature (27), foetal temperatures can rise to dangerous levels if maternal temperatures increase. A systematic review summing 36 studies showed that maternal hyperthermia around the time of childbirth increased odds of neonatal brain injuries, such as cerebral palsy, by 2.5-fold (28). Brain injury in the foetus related to hyperthermia is ascribed to 'energy failure', where the amount of oxygen and nutrients delivered to brain tissue in the foetus are insufficient to meet an increased demand, leading to brain tissue injury and long-term neurological sequalae (29). In those studies, body temperatures above 39°C in the mother had particularly severe consequences for the foetus, but even mildly elevated maternal temperatures (37.1-37.5°C) increased risks of adverse birth outcomes (30).

Secondly, maternal dehydration during a heat event can lead to electrolyte imbalances and, undermine already burdened cardiovascular and renal systems during pregnancy. Within the foetus, dehydration can manifest as severely reduced amniotic fluid during extreme heat events (31)[]. Thirdly, high ambient temperatures can interrupt hormone regulation which can affect the functioning of the placenta, is linked to abnormal glucose tolerance and can affect mental health and wellbeing through increases in adrenaline and stress hormone release. (32, 33). (34). (21, 35). Lastly, heat stress affects placental growth, development and function (36). Chronic temperature stress is associated with reduction in volume and weight of the placenta, as shown in human (36) and animal studies (37). This limits placental transport efficiency, including for oxygen and nutrients. During more acute periods of thermal stress, shunting of blood to the body surface, reduces uterine blood flow and causes reduced placental blood flow (38).

The four pathophysiological processes described above manifest clinically in a broad spectrum of conditions (Figure 1). Maternal health conditions linked with high ambient temperatures include

hypertensive disorders of pregnancy (e.g., preeclampsia and eclampsia) (39, 40), including in a recent study in South Africa (41), preterm pre-labour rupture of membranes (42), placental abruption (43), antenatal and postpartum haemorrhage (39, 40), renal disease (40), and increased rates of emergency Caesarean sections (25, 40). Data are not yet available on the impacts of high ambient temperatures on maternal deaths, though this is likely as heat increases the risk for all the principal causes of maternal deaths. Less frequent causes of these deaths also appear heat sensitive, for example, in one study in the US, the risk of a cardiovascular event during childbirth rose by 11% with each 1°C increase in temperature(44).

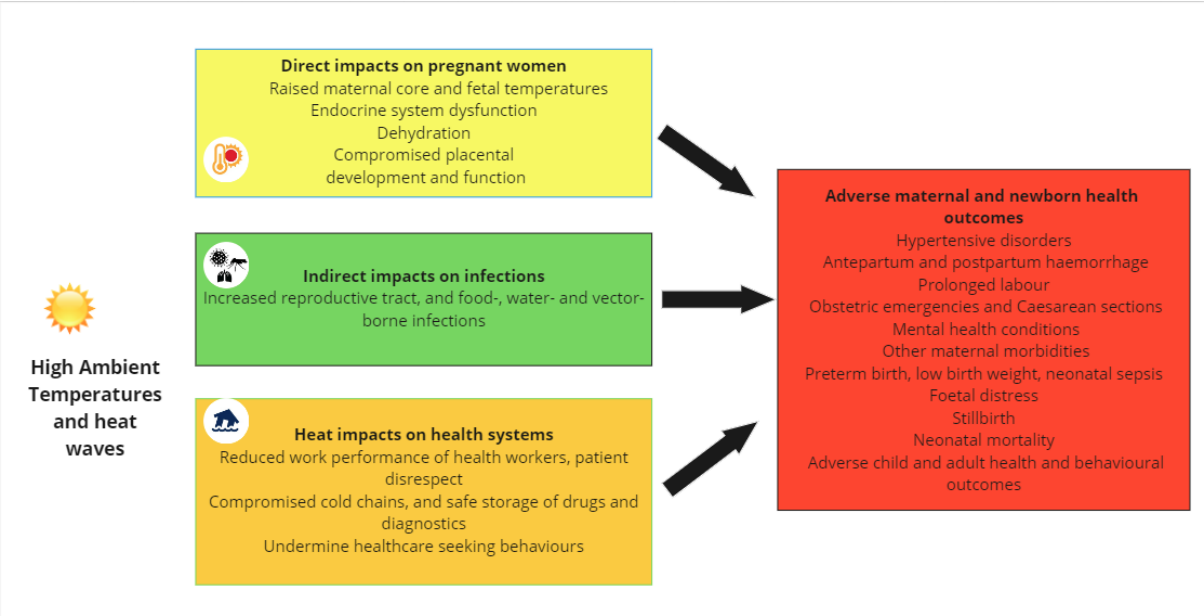


Figure 1: Pathophysiological process of heat, and indirect impacts of high temperatures on maternal and newborn health

Exposure during pregnancy manifests in several ways in the foetus and newborn, but also across all life stages. Heat exposure can lead to increased uterine contractions (45, 46) and foetal heart rate, and reduced foetal movements, as demonstrated in studies among pregnant women exposed to high temperatures in thermal chambers (47), or through intrauterine temperature monitors during labour (48). Clinically, during periods of high ambient temperatures there are increased rates of foetal distress (39), preterm birth (increased in a meta-analysis by 1.16 fold during heatwaves, for example) (49), low birth weight (20), stillbirths (increased in a meta-analysis 1.24 fold at high versus low temperatures) (49), meconium aspiration (39), neonatal jaundice (50), neonatal intensive care admissions (51) and perinatal mortality (increased 1.53 fold in one study when temperature exceeded the 95th centile) (52). Furthermore, several studies, including in Ethiopia (53) and Uganda(54), have shown that exposure to high ambient temperatures *in utero* affects health throughout the life course (55).

2.3 Information gaps

Research on heat and health has mostly been piecemeal, restricted to stand-alone individual studies with relatively small sample sizes, poor-quality data, large variation in research methodology, and limited geospatial and temporal reach. Each additional isolated “small data” study on individual populations and settings offers only limited accruals of knowledge and diminishing returns. Moreover, most studies have insufficient power to answer questions around which specific aspects

of heat exposures and temperature thresholds are most harmful for different clinical conditions, and in which setting and subgroup. Analyses almost exclusively focus on relationships between exposure to extreme heat and common outcomes, such as preterm birth, with little known about heat exposure and rarer outcomes, or composite outcomes where several conditions are clustered together. Using temperature data aggregated to a district or large geographical area as in some studies introduces major measurement biases, as, in reality, there are major variations in temperature across a city, let alone a district. Further, the external validity or generalizability of the findings of these studies is limited beyond the place, and even time period, of the study. Moreover, very little work has been done on heat impacts in some key population groups in Africa, such as pregnant women and children. Rich data resources remain untapped. Africa has seen some of the largest clinical trials globally, there are existing large data repositories with high-quality biomedical data available [63, 64] and many research funders, such as the Wellcome Trust, now mandate data sharing.

Although exposure to extreme heat poses dangers for all people, the burden of heat-related disease falls heavily on specific risk groups (18). Given the unique demographic profile, disease spectrum and built environment in Africa, the most-at-risk groups in these settings will likely differ from elsewhere. On the whole, evidence defining these risk groups is patchy on the continent. An efficient and equitable health sector response to the climate crisis rests on knowing which groups to target and how best to do so. Socio-economic factors in Africa, especially poverty, migration, food and water insecurity, together with high burdens of HIV and other infectious diseases, converge, to form multiple syndemics with climate change, in turn perpetuating vulnerability (56, 57).

In conclusion, only a small proportion of studies on heat-health impacts have been set in low- and middle-income countries, very few of which are in Africa. This is a major gap as these are the very settings which are being hardest hit by climate change, and which have the worst maternal, newborn and child health indices. While people in high-income countries are increasingly turning to air conditioning and mechanical ventilation to provide protection from heat (58), much of Africa lacks the resources to protect people in this way. Put simply, ‘what happens from a clinical, wellbeing and social perspective to a woman in late pregnancy when temperatures reach above 40°C, for example, and she has no access to an air conditioned or ventilated space, no shade and limited water supplies’? And what can be done to protect her?

2.4 Addressing information gaps

Valid quantifications of the current and future burden of heat-health conditions will go some way to highlighting the scale of the threat posed by escalating temperatures to pregnant women and members of other vulnerable groups across Africa. Moreover, as with all health conditions, systems to track changes over time in the overall burden of heat related morbidity and mortality are key to a successful health sector response. We hold that developing and testing analytical systems using maternal and child health data will pave the way to solving the major health challenges described above.

One such analytical system that we propose is an individual participant data (IPD) meta-analysis. An IPD, review, defined by Cochrane, is a “specific type of systematic review that involves the collection, checking and re-analysis of the original data for each participant in each study (59). An IPD includes studies from a large geographical area and manages to secure data from a high proportion of eligible studies and can provide highly robust evidence with considerable external validity or generalisability

(Figure 2). IPD analyses reduce selection bias by drawing on databases regardless of whether the exposure-outcome of interest has been reported, or if information is available on the potential presence or size of the association (60). Individual-level information enables more flexible and robust analyses than are possible with aggregate study results in published data, including the ability to deal appropriately with time-to-event and longitudinal data.

Reuse of primary data from studies and reporting of summary syntheses of these analyses offers considerable advantages over traditional systematic review methods that use aggregate data extracted from publications (Figure 2) (60, 61). The advantages of IPD methodology are especially apparent with systematic reviews of published data on observational studies, such as those on temperature and health outcomes. Those reviews are plagued by the considerable methodological diversity of the included studies, including variability in exposure and outcome measurement; operationalisation of temperature metrics; lag time from exposure to outcome; and indeed the methods of data analysis. Few, if any, of these studies adhere fully to reporting guidelines such as OBSERVE or pre-specify analysis plans. IPDs provide the opportunity to report findings in a fully transparent manner, where it is clear to readers which analyses were pre-specified and which were exploratory, and why.

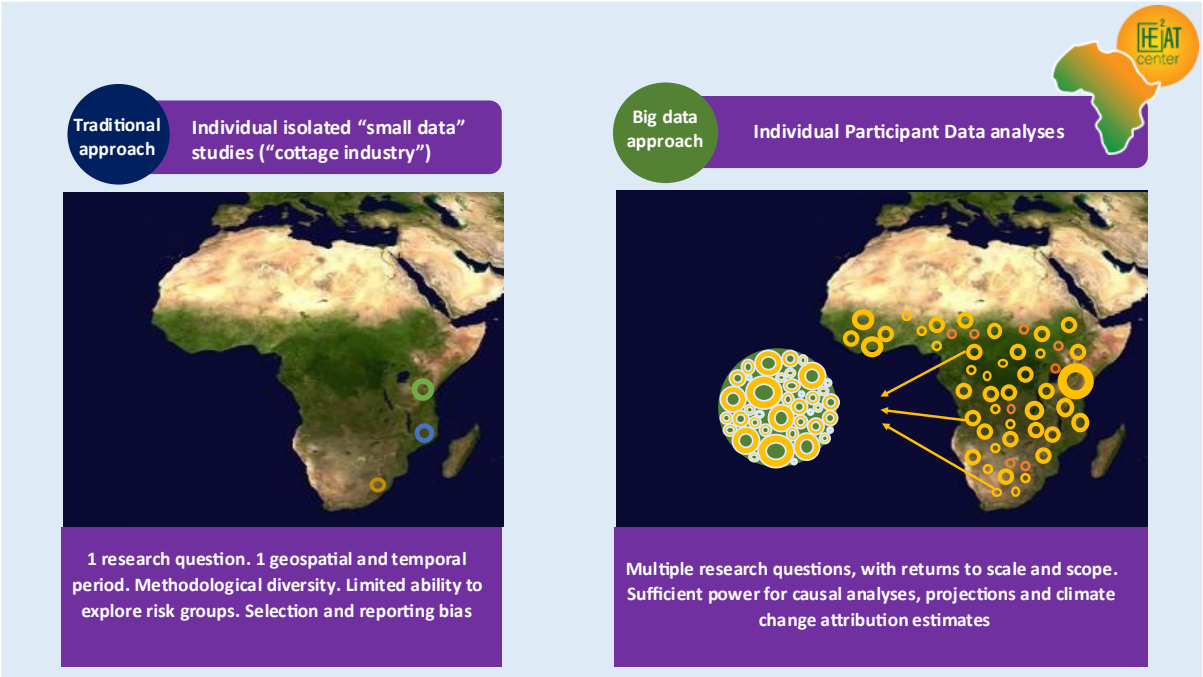


Figure 2 The differences between traditional and IPD analysis approach to heat-health research in sub-Saharan Africa

There are many potential advantages in analysing longitudinal data collected in cohorts or trials, especially where the relationship between the exposure and outcome is highly complex and lagged. Both exposure and outcome measures on individuals may be of higher quality in longitudinal than other data types, have larger numbers of variables – both exposures and outcomes – and allow for more nuanced time-to-event analyses. In particular, such data can allow for analyses that quantify associations on each of the steps in a putative causal pathway (62).In particular, such data can allow for analyses that quantify associations on each of the steps in a putative causal pathway (62).

The end goal of the IPD is to produce overall estimates of impact across the databases assessed. But, equally, variations in findings by population group, setting, time period and climate zone. One key means of identifying interventions, especially those in the public health arena, is to isolate the factors that account for variations in findings between studies. The factors that determine whether a population is affected by a particular exposure, may well be the same factors that an intervention may target, or be built around. Big datasets allow for in-depth exploration of heterogeneity – delineating the reasons for variations between studies – and can generate actionable insights. Explicitly setting out to use data on exposure-outcome associations to gather actionable information to inform intervention design is a novel addition to methodological approaches in this field.

Overall, the IPD approach is relatively rarely used across all research fields, perhaps owing to the time and resources required by a research consortium to systematically identify eligible studies, build collaborations of individual study investigators, and to harmonize the raw data into a common set of variables across the studies. While these initial steps are time-consuming, there are important benefits once the data have been prepared at scale for analysis. Analysis of IPD proceeds along the same principles as analyses of an individual database with the added benefit that data can be dynamically added to the analysis as it becomes available. Moreover, once the data processing and analysis systems are established, real-time tracking or monitoring of exposure-outcome associations and burden of disease can be done. Here, for example, data sources from patient hospital records, can be analysed real-time as they emerge and will require few resources to do so.

2.5 Public health impact of the project

A detailed understanding and quantification of the linkages between climate and health is essential for developing solutions to this critical policy and research area. High-quality, adequately powered 'impact studies' which document the health effects of heat are especially important as these exposures are the principal direct hazards of climate change. By analysing heat impacts using large databases from multiple settings and time periods we aim to achieve three overarching goals.

First, by identifying the specific populations, circumstances and settings which are most vulnerable to heat we can optimise the efficiency and equity of the health sector response through targeting resources to those most in need and provide insights into potential interventions that may be employed. The WHO, other international agencies, and large bilateral donors who have successfully led the response to other major health challenges in Africa, have yet to address the problem of extreme heat exposure on the continent.

Second, by deepening our understanding of the causal pathways between exposure to heat and health outcomes we lay the groundwork for the design of adaptive interventions, including district level surveillance systems. Third, by performing adequately powered and high-quality 'impact studies', we will generate the information required to calculate the burden of disease from climate change which, in turn, strengthens arguments for allocating sufficient resources to address climate-related impacts and for hastening the societal changes required to avert further changes in climate.

The analysis of heat exposure and maternal health linkages we propose will provide a proof of concept for IPD application to address other key questions in climate change and health more broadly. We aim to set the standards and develop a prototype solution for the field, which can then be replicated in analyses of other heat-health or pollution-health associations.

Testing quantitative analytical methods on the health outcomes of pregnant women and children provides critically important information about heat impacts in these priority population groups. The intention of the quantitative analytic approaches to be developed in this project is to provide a level of location and time-based detail that is relevant for informing current policy, clinical practice and guidance of public health responses to the climate crisis. Moreover, while the project will generate important information that is needed now, its full potential and that of the overarching approach will become evident as exposures to extreme temperatures and other climate change manifestations escalate in the coming years.

At the conclusion of the project, we aim to develop and test an indicator of climate change and health to be used in the District Health Information System**.** The protocol does not cover the indicator development, which will be described in a later document. Here, the machine learning algorithms that are constructed in the IPD analyses to identify and assess heat-heath associations will be applied to district-level data. We believe that our proposed indicator of excess adverse health outcomes due to heat exposure will have high validity and be feasible to implement, and, if so, there will be much interest in adopting the indicator across the sub-continent, especially with advances being made in electronic medical records or in ways of converting from paper-based to electronic data.

3. Study objectives

3.1 Primary objective

1. To use innovative data science approaches to quantify the current and future impacts of heat exposure on maternal and child health in sub-Saharan Africa.

3.2 Secondary objectives

Secondary objectives are:

1. To acquire, collate and integrate prospectively collected data from cohort studies and randomized trials on maternal and child health in sub-Saharan Africa
2. To develop a collaboration between study groups and the IPD investigators
3. To link health outcome data spatially and temporally with weather and other environmental data, as well as socio-economic and other data
4. To utilize classic statistical and novel machine learning approaches to understand and quantify the impact of heat exposure on pregnant women and children
5. To document in detail the variations in the relationship between heat exposure, and maternal and children health outcomes across different settings, climate zones and population sub-groups.

3.3 PECOS questions (Population, Exposure, Comparator, Outcome, Study design)

The following criteria using the PECOS framework is outlined below to refine the research question.

Table 1: PECO question for IPD review of maternal and child health outcomes

PECOS	Definition
Population	Women in sub-Saharan Africa in pregnancy, childbirth and up to two years postpartum, and their children up to 2 years of age, exposed to high temperatures*
Exposure	Ambient temperature (and other environmental exposures) at the time of conception, pregnancy and childbirth, analysed in different lag periods
Comparator	Women in sub-Saharan Africa in pregnancy, childbirth and up to two years postpartum, and their children up to 2 years of age, exposed to low temperatures*
Outcomes (primary)	**Maternal outcomes** 1. Hypertensive disorders in pregnancy 2. Antepartum and postpartum haemorrhage 3. Infections in pregnancy or postpartum **Neonatal and child outcomes:** 1. Preterm birth 2. Stillbirth 3. Low birth weight 4. Intrauterine growth restriction 5. Stunting

Outcomes (secondary)	**Maternal outcomes** include duration of labour, premature rupture of membranes, Caesarean section rate, maternal death, gestational diabetes hospital visits/admissions **Neonatal and child outcomes** include small for gestational age, congenital anomalies, foetal distress, hospital visits/admissions, diarrhoea, pneumonia, mortality

*Temperature is defined in a number of ways; we mention high versus low temperature as a simplified means of noting temperature comparisons in the PECOS question. Some comparisons will be degree increments in temperature from minimum to maximum temperature, for example, and thus not ''high'' versus ''low'' temperatures.

3.4 Study hypotheses

3.4.1 Primary hypotheses

1. The rates of preterm birth are increased when maximum temperatures are above the 90th percentile in the week prior to childbirth.
2. Exposure to maximum temperatures above the 90th percentile in the three days before childbirth and during childbirth increases the risk of maternal morbidities.
3. Exposure to a higher number of hot days during pregnancy is associated with intrauterine growth restriction and low birth weight.
4. A higher number of hot day exposures during pregnancy and the first two years of childhood is associated with stunting in children under the age of two.

3.4.2 Secondary hypotheses

1. High ambient temperatures and heatwaves during pregnancy and intrapartum on maternal and child health, vary across key population sub-groups, including maternal age, socio-economic status, geographical area, and climate zone.
2. Exposure to high ambient temperatures increases the burden on health systems due to increased rates of caesarean sections and out- and in-patient health facility visits.
3. High ambient temperature exposure during the first two years of childhood increases the risk of diarrhoea, pneumonia, all-cause out- and in-patient health facility visits, and all-cause mortality.

3.4.3 Rationale for the study hypotheses

The primary hypotheses have been selected to explore important maternal and child health outcomes, varying periods of exposure (during pregnancy, childbirth and after childbirth), as well as varying heat exposures such as maximum temperature above the 90th percentile, high ambient temperatures, and heatwaves.

The maternal outcomes have been selected based on evidence of heat-health impacts presented in the background, as well as in alignment with the top causes of maternal mortality in sub-Saharan Africa. Sub-Saharan African has a disproportionate burden of maternal deaths, accounting for two-thirds of the world's maternal deaths (63). The top five causes of these deaths are severe bleeding, infections, high blood pressure during pregnancy, complications from delivery and unsafe abortion (63). The contribution of heat to these highly prevalent outcomes will be imperative to quantity, to improve maternal health in sub-Saharan Africa. Similarly, the child health outcomes have been selected based on the most common causes of morbidity and mortality in that period. Almost one in

two deaths in children under the age of 5 occur in the neonatal period, with sub-Saharan Africa having the highest rates of neonatal mortality. The main causes of death in the neonatal period are preterm birth, intrapartum-related complications, and infections (64, 65). Following the neonatal period to the age of five, the most common causes of death are pneumonia, diarrhoea, birth defects and malaria (65). Many of these outcomes have been selected in the primary and secondary hypotheses, as contribution of heat to these outcomes needs to be assessed. For the secondary hypotheses, appropriate subgroups have been selected based on the PROGRESS-Plus acronym, which is used to identify variables or characteristics that stratify health outcomes (66, 67). The PROGRESS-Plus acronym is as follows: **P**lace of residence, **R**ace/ethnicity/culture/language, **O**ccupation, **G**ender/sex, **Re**ligion, **S**ocial capital, **S**ocioeconomic position, age, disability, sexual orientation, other vulnerable groups.

Varying temperature variables have been selected in the primary and secondary hypotheses. Specific variables have been chosen for the primary hypotheses, for example maximum temperatures above the 90th percentile, while broader definitions such as high ambient temperature have been selected for the secondary analysis. The patterns of temperature and health outcomes are mostly U-shaped with the base of the curves commonly at relatively mild temperatures around 20°C, with curves becoming progressively or exponentially steeper with each increase in temperature degree. Exposures were chosen to encompass the effects of high temperatures that are relatively milder or closer to the bottom of the curve, as well as hot days and consecutive days of hot temperatures in a heat wave, at the top of the U-shaped curve.

3.5 Funding acknowledgement

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4. Individual Participant Data Methodology

An IPD meta-analysis will be conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) IPD extension guidelines (68). The PRISMA statement was designed principally for systematic reviews and meta-analyses of randomized trials that use aggregated data from published articles. A PRISMA statement 'extension' was developed to cover methodological considerations unique to the IPD approach (68).

An IPD entails systematically locating, collating, appraising, transforming, analysing and synthesising data from multiple studies which have a common exposure and/or outcome of interest (68). Therefore, unlike classical systematic reviews, IPD analyses do not rely on aggregate data extracted from journal publications. Rather, the original source data from cohorts or randomized trials are sought and reanalysed. The word "Participant" is used to describe the unit of analysis, which is an individual person, either at one point in time or at multiple points over time in longitudinal studies.

The PRISMA flow diagram extension of IPDs (Annex 1) will be used to illustrate the procedures undertaken in this study. Additionally, the PRISMA-IPD Checklist (Annex 2) is a tool that is used to guide the design and reporting of this IPD.

4.1 Protocol registration and previous outputs

The methods and findings of the MASCOT-1 systematic review was reported in several articles (69-73) and used to inform WHO guidelines on maternal health. The review protocol for MASCOT-2, the systematic search and identification of eligible studies, is registered in PROSPERO [PROSPERO 2020 CRD42020214637] and is available at: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020214637. This protocol has been registered on PROSPERO CRD42022346068

4.2 Eligibility criteria

Eligibility is determined at the study level (where a particular study needs to fulfil the criteria defined below in section 4.2.1) and at an individual level, where a set of criteria determine the eligibility of individual participants within the eligible studies (section 4.2.2).

4.2.1 Eligibility criteria at the study level

The IPD includes studies that collected longitudinal data from medium-to-large cohorts and clinical trials. A cohort, defined by Cochrane is, "an observational study in which a defined group of people (the cohort) is followed over time. The outcomes of people in subsets of this cohort are compared, to examine people who were exposed or not exposed (or exposed at different levels) to a particular intervention or other factor of interest." (1) The National Institute of Health (NIH) defines a clinical trial as "a research study in which one or more human subjects are prospectively assigned to one or more interventions (which may include placebo or other control) to evaluate the effects of those interventions on health-related biomedical or behavioural outcomes."(2) Longitudinal data allows for the assessment of temporal trends and the use of statistical approaches like time-to-event analyses which are valuable in assessing heat-health associations. Including data from longitudinal studies, where women are followed over pregnancy in one location, may avoid biases in birth

registries where a women may have given birth in a place that is some distance away from where she may have spent much of her pregnancy.

The study only includes cohorts/trials that enrolled more than 1000 participants given that the large amount of time and resources required for data acquisition, preparation, and analysis of each individual study, makes it difficult to justify the inclusion of smaller studies.

The following criteria are required for a **study to be eligible for inclusion**:

1. Cohort or trial that enrolled at least 1000 pregnant women in a study site(s) in sub-Saharan Africa, published between January 2012 and 30 June 2022
2. Located through published literature in the systematic mapping, clinical trial registry or other means
3. The study can be single- or multi-site, and across one or more countries in Africa, or other parts of the world
4. Study design is a randomized or non-randomised clinical trial, or an observational or interventional cohort with prospectively collected data
5. At least one primary outcome or two of the 'important' maternal and neonatal/child have been collected as part of the study (Annex 4)
6. The study had obtained the relevant local ethics approvals and any concerns around ethical procedures have been clarified by the data provider

The following are **exclusion criteria** at the study level:

- Systematic and narrative reviews, reports of birth registries, cross-sectional studies, health and demographic surveillance data, or case control studies

4.2.2 Eligibility criteria at the participant level

Even though a study may fulfil the eligibility criteria, some individuals within the study may be ineligible. We apply a set of eligibility criteria for individuals within a study.

The following criteria need to be met for a ***participant (woman and her child)*** to be eligible for inclusion

1. Be a participant from an eligible study
2. The woman was enrolled into the study either during pregnancy or intrapartum
3. Individual participant information is at least available on date of birth of the newborn, or date of end of pregnancy in cases of maternal deaths or abortion
4. Individual participant information is at least available on location of birth, or location of study follow-up (city-level or more granular)

4.3 Identifying studies – information sources

Studies will be located in several ways. Firstly, we will draw on studies identified through online database searches for two systematic reviews. The first review, MASCOT-1 was conducted in 2012, and the following electronic sources were searched: Medline (PubMed), CINAHL, Embase, LILACS, PopLINE, PsycINFO and Web of Knowledge. The second review, MASCOT-2, was an update of this original search and was conducted in 2020 in Medline (PubMed).

Secondly, we search through data repositories, such as the Bill and Melinda Gate Foundation Knowledge Integration-platform, National Institute of Child Health and Human Development Data and Specimen Hub, ClinicalTrials.gov and other relevant research data storage initiatives. Lastly, we will seek additional studies through direct contact with study investigators of the individual cohorts/trials identified or other experts in the field. These last two processes are particularly important for identifying data not published at the time that the search was done.

4.4 Identifying studies - search strategy

MASCOT-1 used both controlled vocabulary and free-text terms to identify studies on PubMed. This search strategy was then adapted for subsequent searches of other electronic databases. MASCOT-1 included search terms for maternal health, and World Bank defined (2012) low- and middle-income countries and was limited to the period from 2000 to 2012 (Annex 3) (74). Terms for maternal health covered the antenatal and postnatal periods, as well as outcomes including biological, process, or health systems outcomes. The items located were screened in duplicate, independently, and categorised by study design, setting and other variables. For our IPD study, longitudinal studies conducted in sub-Saharan African countries were then filtered out from studies within the full MASCOT-1 review.

MASCOT-2 (Annex 3) used the same search terms for maternal health, but only included World Bank defined (2019) sub-Saharan African countries as well as validated search filters for locating cohorts and clinical trials. The search filters to identify cohorts and trials was taken from the library guides from the School of Public Health at the University of Texas (75). The search terms for maternal health, cohort/trial study type, and sub-Saharan African countries were then combined for the full search. For MASCOT-2, we searched MEDLINE (PubMed) database for articles in the period from 1 September 2012 to 10 October 2020. We also searched the Cochrane clinical trials register for additional studies.

The full search strategy for MASCOT-1 and MASCOT-2 in PubMed is included as Annex 3.

4.5 Study selection processes

Using EPPI-Reviewer software (76), screening of titles and abstracts for potential eligibility was done independently, in duplicate, with differences between reviewers reconciled through discussion, or by a third reviewer. If potential eligibility could not be ascertained from the title or abstract, then the full text was also screened. Publications of studies were considered for potential inclusion if on screening on title, abstract or full text, the inclusion/exclusion criteria described in section 4.2.1 were met.

4.5.1 Extraction of data from full text articles

The full text articles for potentially eligible studies were sourced and if located, uploaded onto EPPI-Reviewer for data extraction. Extraction was done individually, in duplicate.

The following variables were extracted during this process:

- Study population: sub-Saharan African country of study (as defined by the World Bank), number enrolled for the whole study as well as the number enrolled in each sub-Saharan African country for multi-country studies

- Study methods: study design, intervention (if applicable)
- Study Identifiers: year of publication, study name, study acronym, clinical registration number, authors, study funders
- Data availability statement if available

We checked clinical trial registries for additional information that had not been provided in the full-text published article.

4.5.2 Deduplication of eligible studies

One or more publications may be linked to a single study. Consequently, an additional step to deduplicate publications that are linked to one study, is required. In most instances, this was done using the study name, clinical registration number, authors/PI details, enrolment numbers or other reported study identifiers. The Pubmed "LinkOut" function to the ClinicalTrials registry was also used to help link and group publications together. Where two or more publications contain contradicting data, for example, a different sample size, but other details such as PI name or methodology are the same, further investigations are warranted. We will search through the registration details on clinical trial registries such as ClinicalTrials.gov and WHO International Clinical Trials Registry Platform, to ascertain the registered study name, registered PI details and enrolment number to resolve these differences. In some cases, there may still be inadequate information to determine whether a publication is drawn from a particular study, in which case, the PI or first author listed on the paper will be contacted for clarification.

4.6 Data acquisition processes

Data will be acquired either through data access platforms that store and curate research data for further re-use, such as Worldwide Antimalarial Resistance Network (WWARN) (77), or directly from study PIs/other relevant study custodian, who will hereon be referred to as the data provider. Data acquired through these data access platforms will oftentimes require a data request form that provides further details about the intended study. The data request form is then reviewed by a data access committee before a decision is made on granting access to the requester. We will complete and submit these data request forms as required and if granted access to the data, we will complete a formal data sharing agreement.

In a pilot phase prior to the submission of the grant proposal for this study, we gauged the interest and wiliness of data providers to share their data for the purpose of this IPD through an email request and had very positive responses. Similarly, once we have identified studies to be included, we will send an email request to the corresponding author(s) of those articles, which will detail the purpose of the study and outline the collaboration. Thereafter, should we have no response from the corresponding author, we will contact the first or last author, or other authors listed. We will try at least five times to make contact, and explore other means of making contact, such as LinkedIn. We will document the number of attempts made at contact and modes of contact.

Data providers that agree to joining the collaboration will sign a data sharing agreement which sets out the terms of data sharing, data security and authorship guidelines. In addition, a series of ethical checks, and approvals if required, will be conducted, as outlined in section 7 of the protocol.

Collaborators will supply meta-data, or key documentation for their study. This will include the full study protocol, informed consent forms, codebook, details of the ethics approval of the study, and will be contacted for additional information if required. We will periodically update the data providers on progress with the study and involve them in the data analysis, article writing or other aspects of the study where possible. We will have frequent email exchanges, and face-to-face or online meetings with the data providers where practicable and use these opportunities to discuss data queries if required.

We will record the process of trying to obtain IPD data, and, if applicable, any reasons for being unable to obtain IPD data in the flowchart, such as inability to contact data providers, investigators unwilling or unable to contribute their data, or data from the identified study has been destroyed.

4.6.1 Environmental data acquisition

Climate data include both observational based datasets (weather station observations, satellite proxy observations) and processed/gridded observations. Gridded climate data produced from atmospheric re-analysis and climate simulations will form historical gridded climate observations and forecasts. Climate related data will in almost all cases involve accessing open data repositories such as Copernicus Climate Data Store (CDS) or Earth System Grid Federation data systems. Climate related data will either be - stored on IBM Physical Analytics Integrated Data and Repository Services (PAIRS) data storage and/or Climate Science Analysis Group/University of Cape Town (CSAG/UCT) data storage systems. The IBM PAIRS data can be made available through the CSAG/UCT Jupyter Hub platform through an API or queried directly from the IBM PAIRS platform. CSAG/UCT will manage and update the primary data index for climate related data available to the consortium. All climate datasets that will be used are available through open data policies with no restrictions on non-commercial research use.

Remotely sensed data from satellite sensors, mainly optical imagery (e.g. satellite images of urban centers) are a source of valuable information about physical attributes such as land surface temperature, soil moisture estimates, vegetation condition, land use and cover and air quality, etc. Raw and processed geospatial datasets from satellites will in almost all cases involve accessing open data repositories such as Copernicus Climate Data Store (CDS) and sentinel data systems. There are large volumes of these data downloaded from these open data repositories that have already been pre-processed that are currently stored on the IBM PAIRS and the CSAG/UCT data platforms. Where data sharing licenses permit, these that will be available for use to partners and outside groups, together with other open remote sensed geospatial datasets that have been downloaded and stored on the CSAG/UCT data storage systems. CSAG/UCT will manage and update the primary data index for remote sensing related data. All remote datasets to be used are those available through open data policies with no restrictions on non-commercial research use.

Air pollution is a major exposure of concern when it comes to the specific maternal and child health outcomes being investigated in this study. The finding from a review of several studies on US population conducted between 2007 and 2019, was that exposure to poor air quality (high levels of $PM_{2.5,}$ ozone and other pollutants) was associated with higher risk of pre-term birth, low birth weight and still birth (78). These same adverse obstetrical outcomes were also associated with exposure to heat extremes. There is thus an impetus to investigate the exact nature of the combined impact of exposure to air pollution and heat on maternal health outcomes.

Air pollution data will be required to consider the joint exposure of participants to both air pollution and heat. The challenge to air pollution analyses in Africa is gaps in the coverage of the ground-based air quality monitoring network of stations as shown in Figure 3. Therefore, proxy satellite-derived air quality data such as Aerosol Optical Depth (AOD) from for example Sentinel 3 or the Moderate Resolution Imaging Spectroradiometer (MODIS) onboard the Earth Observing System Terra and Aqua in combination with land cover and use data can be another option to consider for the provision of estimates of air quality at each of the study locations (79). Given the nature of satellite AOD data (i.e., differences between these data and the ground observations in spatial resolution and temporal coverage, among other issues) the estimation of air quality at the various locations of interest in Africa will be a significant task. Ideally data on pollutant concentrations, namely of PM_{10}, $PM_{2.5}$, NO_2, SO_2 and CO, at a minimum can essentially be used in developing indices of air quality. Sources of historical databases with global coverage include, the World Air Quality Index (80), where data has been collated from 2012 and the index is determined using the US Environmental Protection Agency standards. The OpenAQ (81), an open data platform, provides downloads of historical and near real-time (harmonised) physical air quality data (not indices). The historical coverage of the data from the OpenAQ is likely to conform with the temporal coverage of the air quality monitoring campaign in each country. Data from these sources will be downloaded onto the CSAG/UCT platform for storage and processed for missing data and other data quality concerns prior to the creation of an "analysis-ready" air quality dataset that will be available to merge with the individual participant data frame. Another source of air quality data which may be used for checking estimates, especially in countries where there may be very little to no stations is from the WHO and it consists of country level population weighted annual mean concentrations of PM_{10}, $PM_{2.5}$ (82), available for several years (not all) since 2010.



Figure 3: Real-time air quality index for PM2.5 globally. The map shows the coverage of the monitoring network in Africa (83)

4.7 Data items

4.7.1 Study level variables

For each study, we will collect information on key study and participant characteristics, such as study aims, information on interventions if relevant, total numbers of participants enrolled, demographic data, health outcomes, funding source, duration of follow-up, and location of study (Table 2). This information will be extracted from published study articles identified through the search strategy.

For unpublished studies, investigators will be asked for a study protocol which should entail the study level variables.

Table 2: Study level variables

Study variables
Study design
Aims
Location of study
Type of intervention (if any)
Duration
Participant recruitment dates
Number enrolled
Participant baseline demographics
Outcomes
Funder
Study source (e.g. systematic review, personal contact, data repository)

4.7.2 Participant level variables

Maternal, neonatal and child health outcomes, and climate variables are detailed in Annex 4. The outcomes have been divided into essential, important and desirable, based on the primary and secondary hypotheses in section 3.4, and important maternal and child health outcomes in sub-Saharan Africa. The health variables will be defined using published, reputable sources such as the WHO, the Global Alignment of Immunization safety Assessment in pregnancy (GAIA) terminology (84), and/or local obstetrics and paediatric guidelines when used in the IPD. Each participant will also have a set of climate variables, including heat indices, which are further detailed in Annex 4.

5. Data analysis plan

The analysis plan outlines various analysis steps that will be undertaken, culminating into a two-stage IPD meta-analysis, that will be implemented with data that has been cleaned, harmonised and organised. In figure 4, we illustrate the broad steps that will result in the comprehensive database that will be used. In the first phase, the focus is on the collection of data and metadata (including data dictionaries) for studies to be included in the meta-analysis. In particular, characteristics of the data relating to its quality or integrity will be noted during this phase. This will inform the pre-processing required to address issues such as missingness as well as variable selection. The second phase, which will run parallel to the first, will consist of pre-processing both the climate and environmental data, with the aim of deriving variables that will be included into the harmonised health database, resulting in an integrated individual participant database consisting of clinical health outcomes variables, demographic and health covariates as well as climate and environmental covariates.

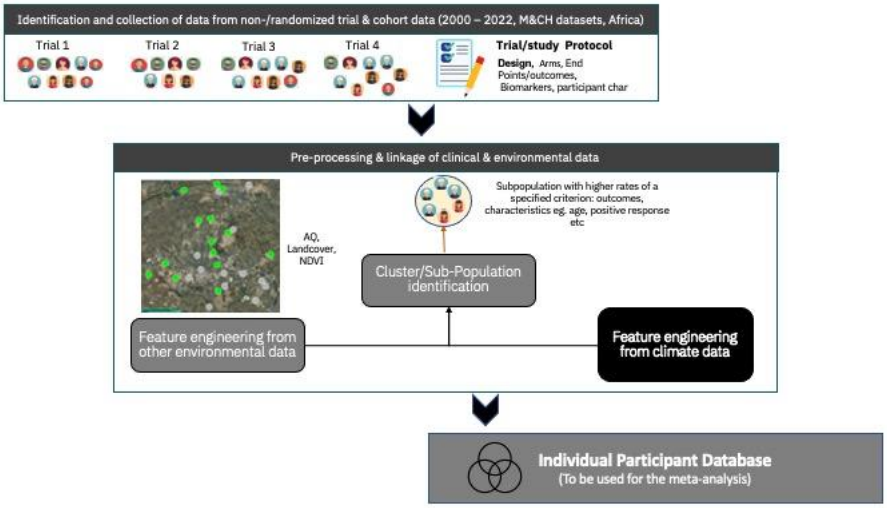


Figure 4: Two-stage IPD meta-analysis, including the linkage of environmental data

5.1 Individual participant data integrity

The IPD will be subject to data checking for completeness and integrity using various strategies. Each dataset received will be checked to ensure that the data can be opened, and that it is the correct dataset for the study. Consistency of the data will be checked by comparing the dataset to relevant information from study publications like number of participants enrolled and baseline demographics. Additionally, more in-depth checks will be conducted on a smaller subset of the dataset to look for missing, invalid, inconsistent, or out-of-range values, for example, unusually aged pregnant women, out-of-range blood test results, and visit dates before trial start date (85). The data provider of the study will be contacted to resolve any discrepancies found.

5.2 Risk of bias assessment in individual studies

Quality of the studies will be evaluated in a risk of bias assessment, using tools based on the study design. The assessment will be done by two investigators independently, and the queries resolved

through discussion, and if necessary, consultation with a third investigator. The table below describes the tool that will be used for each study type (Table 3) and the tools are available in Annex 5. The most updated version of the tool will be used at the time of assessment.

Table 3: Risk of bias tool for different study designs

Study Design	Risk of Bias Assessment Tool
Randomized controlled trial	Cochrane risk-of-bias tool for randomized trials (RoB 2) (86)
Non-randomised studies of interventions (non-randomized clinical trials, cohort studies)	Risk Of Bias in Non-randomized Studies of Interventions (ROBINS-I) (87)

Each study and study outcome will have an overall grading which will be taken into consideration in meta-analysis and sensitivity analyses.

5.3 Synthesis methods

A two-stage analysis approach will be used primarily, whereby, in the first-stage, each study is analysed individually. In the second stage, the data from the individual studies are aggregated to provide an overall pooled estimate of effect. It is important to note that a one-stage analysis approach may be needed and appropriate in some circumstances. It is also possible that the confidence intervals around an estimate in individual studies is very wide. In these instances, then having a two-stage approach may not be optimal, but rather a one-stage approach, but with the final model to include the variable study identification. Additionally, if there are low levels of heterogeneity across the studies, then a one-stage approach is an appropriate method. In some instances, we may present both a one- and two-stage approach.

Once a dataset has been checked, temperature, air pollution, other environmental, socioeconomic data and date of the health event (e.g. date of birth, or date of adverse health event) are merged with the database. The meteorological data are linked to time and place in each study, aiming for small geographical units, wherever possible. Baseline characteristics of participants from each of the participating cohorts or trials will be described.

5.3.1 Systematic data preparation and quality assessment

Using data from multiple trials through an IPD offers a powerful means of controlling for confounding and investigating the size and shape of individual-level effects. The challenge, however, is that data from multiple studies will be characterised by differences in quality that must be addressed, if possible, prior to the synthesis phase of the meta-analysis. Systematic data quality assessment and remediation guidelines that are specific to IPD meta-analyses have been developed and will be used in this study. (88) These have been formulated into a system of five phases of data preparation for IPD meta-analysis, namely, Processing, Replication, Imputation, Merging, and Evaluation, also called PRIME-IPD. The items or data preparation tasks considered under each phase are described in Table 6.

Table 4: Checklist for PRIME-IPD tool (89)

PRIME:	Items
Processing	• Convert data into a single format for statistical program of choice • Compare the total number of participants in the acquired datasets to those reported in published studies • Verify the presence of the variables of interest in the acquired dataset • Standardize variable names across datasets • Identify and standardize the measurement scales used to report the variables of interest • Identify and standardize coding for missing values • Identify and correct any implausible values that may result from data conversion
Replication	• Recalculate reported descriptive and summary statistics using the acquired datasets • Calculate the standardized difference to quantitatively assess the difference between the replicated and published results • If the standardized difference is $> 10\%$, investigate and address potential causes
Imputation	• Assess the appropriateness of conducting imputation of missing data using missing data theory • If multiple imputation is conducted, carefully consider the number of imputations to be run
Merging	• Ensure in processing step that variable order and codes are correct • Merge the imputed datasets into a single, pooled dataset, taking into consideration the number of imputed datasets, if appropriate
Evaluation	• Assess continuous variables for normality by residual analysis either visually or by statistical tests • If required, calculate new variables for standardized comparison of effects

Processing

Much of the processing phase is focused at translating the raw, disparate datasets from the multiple trials into a harmonised dataset. The data harmonisation process is described in further detail in the data management (section 6) and the data management plan (Annex 6).

Replication

Standard exploratory data analysis techniques will be used for this step or phase. Here the aim is to reproduce the summary statistics reported for each study in published articles, assessing for significant deviations to what was reported. The criterion to be used for indicating significance of the deviation is based on the standardized difference, where values greater than 10% will indicate that further investigation is required for that particular dataset. We may not be able to replicate the published evidence on the association between heat exposure and health outcome as there are very few such studies in Africa. We will, however, be able to perform this replication steps on individual variables such as age, where we compare the age of participants as reported in the article with the age in our analysis.

Imputation of missing data

Missing data, especially when the proportion of missing data is significant, pose a challenge in any study. Depending on the pattern of missingness (i.e., random vs non-random) estimates may be affected in several ways. Non-random missingness and use of complete case analysis when the proportion of missing data is high (>50%) can lead to biased estimates. The presence of missing data can also lead to increased uncertainty. There are several approaches to imputing missing values, including the naive mean imputation and complete cases analysis which may be used in this study for those variables where the assumption of missing at random and the proportion of missing data is very low. Upfront it can be noted that participants will be excluded if date of birth of the newborn and location data are missing.

For the most part, it is envisioned that for a complex study such as the planned meta-analysis, it is appropriate to consider model-based methods to imputation, including imputing missing data based on data from other studies in the IPD with careful attention to congeniality (same assumptions, e.g., distributional assumptions) between the imputation and analysis model and avoidance of model misspecification. Multiple imputation methods, also known as Rubin's Multiple Imputation methods

(90), including multiple imputation through chained equations (or sequential regressions) which fall into the category of full conditional specification approach, as opposed to the joint distributional modelling approach, are plausible approaches and have been applied in the context of IPD meta-analyses (62, 91, 92). In multiple imputation, multiple complete datasets are generated from sampling multiple times the missing values from the appropriate predictive distribution, conditional on the observed data. The substantive model of interest is then fitted to each imputed dataset, and the results combined for final inference using Rubin's combination rules. Uncertainty estimates are determined either by considering a Bayesian or by adopting bootstrap approaches (93). Both choices will be considered for this study, with the eventual choice being informed by the computational efficiency between the two approaches. There are open-source libraries that can be used in imputing missing values, for example the *mice* package for multiple chained equations approach (94) and proprietary software such as STATA have built-in procedures.

Evaluation

This step consists of evaluating the merged dataset's heterogeneity and distributional assumptions applicable to each variable. This is important information that informs the statistical analysis of the pooled data. Visual analytics and formal diagnostic tests will be used to assess distributional assumptions of variables. Heterogeneity among data from the various datasets can also be tested using I^2 statistic, the formula presented below:

$$I^2 = \left(\frac{Q - df}{Q}\right) \times 100\%$$ where Q is the chi-squared statistic and *df* represents the degrees of freedom (95).

The statistic, I^2 is described as a percentage (with confidence interval) of the variability in effect estimates that are due to heterogeneity, rather than by chance (95). A higher percentage represents more heterogeneity across studies. If significant heterogeneity is identified, the data will be explored further to assess the strength of the evidence of heterogeneity, and with the use of statical methodology such a sub-group analyses and meta-regression (95). For example, meta-regression can be used to identify what area-level factors might explain the heterogeneity across settings.

Another activity associated with this component/ phase is the creation of new variables, such as dummy variables, transformations of indicators such as body mass index, created by combining weight and height values.

5.3.2 Statistical methods for the synthesis stage

The synthesis stage of the IPD meta-analysis follows after the data has been prepared and the integrated dataset is ready for analysis. Either a one-stage or two-stage model for the meta-analysis will be chosen. In the one stage model, give that longitudinal trial and cohort data is considered, a classical longitudinal fixed and/or random effects model can be assumed. The objective being to estimate the parameters/ coefficients which represent the overall effect size associated with each individual covariate considered. The inter and intra-study variability would be captured by the covariance model. The one-stage model can however be complex to specify, especially in the case where we will consider environmental exposures over and above the demographic and health covariates. The two-stage model therefore differs from the one stage model in that the longitudinal fixed and/or random effects model is specified, deriving effect estimates at the level/ corresponding to each individual study and then the meta-integration step when the overall effect size is obtained

through a meta-regression of the individual study parameter estimates or alternatively through inverse-variance weighting

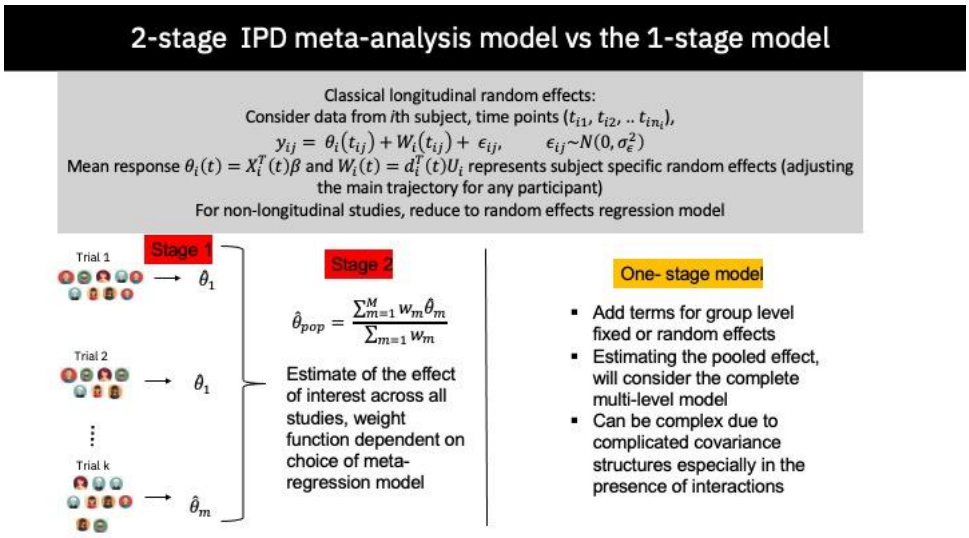


Figure 5: Two-versus one-stage IPD meta-analysis model

To simplify, we can imagine a situation where the longitudinal aspect of our particular individual participant dataset is ignored, and we consider simply that we have participants indexed by *i* from studies indexed by *s* and that we are interested in the estimated coefficients indexed by *j (91)*.

The two-stage fixed effects model can be expressed as follows:

1. Within each study *s*, simultaneously regress $y_{i,s}$ on $x_{1,i,s}$ and $x_{2,i,s}$, obtaining estimates $(\hat{\beta}_{j,s}, \hat{s}_{j,s})$(where $\hat{s}_{j,s}$ is the estimated standard error of $\hat{\beta}_{j,s}$), for *j* = 1,2;
2. Fit the fixed effects meta-analysis model, for *j* = 1,2

$$\hat{\beta}_{j,s} = \beta_{fixed,j} + \hat{s}_{j,s}\epsilon_s, \quad \epsilon_s \sim \mathcal{N}(0,1),$$

The two-stage random effects model

1. Within each study *s*, simultaneously regress $y_{i,s}$ on $x_{1,i,s}$ and $x_{2,i,s}$, obtaining estimates $(\hat{\beta}_{j,s}, \hat{s}_{j,s})$(where $\hat{s}_{j,s}$ is the estimated standard error of $\hat{\beta}_{j,s}$), for *j* = 1,2;
2. Fit the random effects meta-analysis model, using the DerSimonian/Laird estimate of between study heterogeneity

$$\hat{\beta}_{j,s} = \beta_{random,j} + u_{j,s} + \hat{s}_{j,s}\epsilon_s, \quad \epsilon_s \sim \mathcal{N}(0,1); \quad u_{j,s} \sim \mathcal{N}\left(0, \tau_j^2\right)$$

Primary hypothesis

For this study several primary and secondary hypotheses have been specified. For discussing the modelling decisions, we focus on two primary hypothesis that will enable explicit discussion model changes to implement, depending on whether the outcome of interest is binary, time-to-event or continuous variable. The primary hypotheses we focus on are the following:

1. The rates of preterm birth are increased when maximum temperatures are above the 90th percentile in the week prior to childbirth.

2. Exposure to a higher number of hot days during pregnancy is associated with intrauterine growth restriction and low birth weight.

Derivation of the heat exposure variable (effect of interest)

A method to produce higher resolution daily temperature estimates will be developed using a combination of satellite data, ERA5 Land daily temperatures and weather station data. In this context, further advances based on the exploitation of recent developments in geospatial artificial intelligence (geoAI) will be utilised. Some of the approaches that the team will implement include natural gradient boosting algorithm (96), quantile random forest spatial interpolations (97), with implementation of maximum covariance analysis (98) to detect the structure of the covariance between these various forms of spatiotemporal datasets available for the task of producing high spatial resolution temperature data.

- For each higher resolution grid cell, the distribution of the estimated temperature will be the basis upon-which the 90th percentile of the estimated daily temperatures over that cell will be identified
- Each study participant's location is allocated the coordinates of a corresponding grid cell for the climate data
- The participant's date of delivery is used to identify the week prior to delivery
    - A binary variable is developed indicating exceedance during the week before delivery of the 90th percentile for daily temperatures over that particular grid
    - Count of exceedances during week prior to delivery is determined
    - Average daily temperature on days of exceedance
- The three temperature exceedance variables will be appended to the participant data
- Using information on the date of delivery and the gestational age at delivery, for each individual participant, the date range for the pregnancy will be determined. Using this information, the daily temperature values for the grid cell relevant to that individual will be targeted for extracting the number of hot days during the term of the individual's pregnancy. The formal definition for identifying hot days in meteorology will be used.

Health outcome/ response variable assignments

The outcome or response variable can be defined either as a binary variable or continuous variables like birth weight and time-to-event like the time to delivery

i. Binary variable for indicating whether participant had preterm birth or not
ii. Time to event: time to delivery for each individual patient
iii. Continuous variable being birth weight of the neonate at delivery

Machine learning informed covariate selection

For this IPD meta-analysis, data from over 200 individual randomized control trials and cohorts will be collected. The expectation is that the number of covariates available for each study will be large. Further, there is an expectation that the covariates may differ across the various studies. The aim of the harmonisation step, as explained in the data management and analysis plan will be identifying these discrepancies and formulating a strategy, mainly the creation of proxy covariates, to compensate for the different studies having different covariates. Here we give an example with a very limited subset of covariates.

- Age
- Period of birth (month/ seasons)
- Type of delivery

- Smoking (y/n)
- Other covariates

There are traditional approaches to conduct variable selection in the regression context such as forward, backward and stepwise covariate selection. While this is an option that can be pursued, we look to use tree-based ensemble learning approaches, namely extreme gradient boosted trees or random forest algorithms. For both approaches there is a split of the dataset into a training and testing set, the implementation of the tree algorithm and the evaluation of the variable importance ranking and the possibility to use partial dependence plots to identify the functional form between pair-wise or multiple covariates (including interactions) and the response variable. Both forms of tree algorithms can be used for binary, continuous and time-to-event response variables. While autoencoding algorithms may be implemented for feature engineering from the geospatial or spatiotemporal covariate datasets (climate & environmental), ensemble tree algorithms may be used for automatic feature (covariate) selection, maintaining a level of explainability required when trying to understand the health effects of environmental exposures (100). In figure 6, we present an example of an implementation framework in the case of a binary outcome variable, and the ability to develop a logistic regression model with the underlying embedding given by an implementation of a gradient boosted tree algorithm. The same can be implemented for continuous and time to event outcomes using the Gaussian and Cox-regression models respectively.

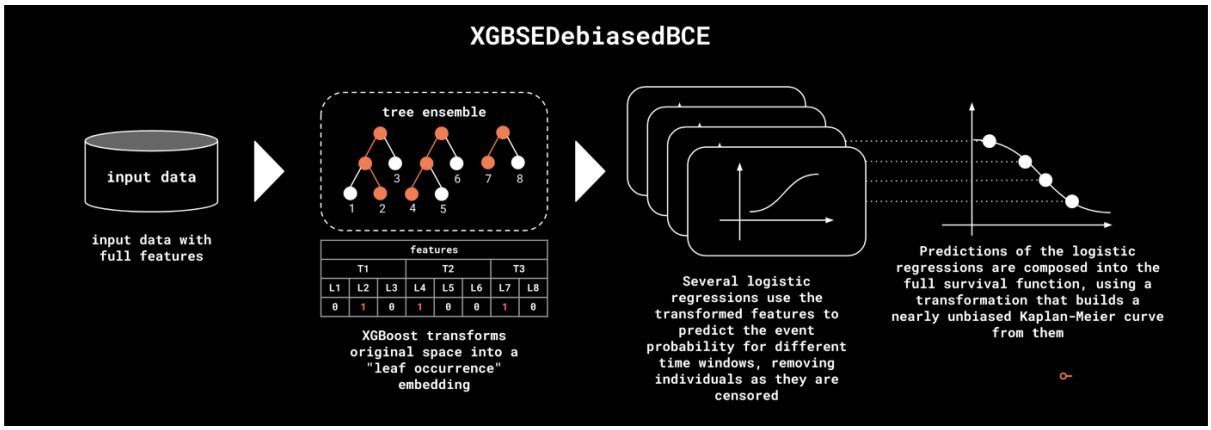


Figure 6: An example called XGBSEDbiasedBCE where several logistic regressions are implemented on top of XGBoost algorithm, where the XGBost is used as a feature transformer and for ranking of the importance covariates. Each logistic regression predicts survival at different user-defined time windows. The predictions are then composed to form a survival (99)

Statistical method for the first stage of the meta-analysis

Statistical methods for detecting exposure-response associations, namely, getting the coefficient estimates, have proliferated in the longitudinal or time series environmental health modelling context (mainly air pollution studies). The core modelling method in this context is linear and non-linear lag-distributed models (95). These are specifically relevant where the outcome variable is a time series, as is typical with hospital counts of adverse events such as preterm births. They have evolved in recent years to incorporate flexibility in the response-covariate structure. The most valuable characteristics of advanced forms of these models, such as the semi-parametric generalized additive model (GAM) following a quasi-Poisson distribution with distributed lag non-linear model used by Mohammadi et al (2019) (96) to model the effects of temperature on preterm births, is the ability to account for non-linear, short-term and lagged effects of environmental exposures on health outcomes of interest. The semi-parametric generalized additive modelling framework, in

general, provides the flexibility to account for non-linearity as well as over-dispersions in the temporal dimension and clustering in the spatial dimension which is pertinent in the case where the outcome of interest is a spatial timeseries of counts. The additive modelling framework may be expanded further to account for multiple health outcomes and associated uncertainties (97), as well as spatial effects and interactions (98, 99). Therefore, large geospatial (and spatiotemporal) data from satellites and sensor networks, for example, can be leveraged as additional features in testing environmental exposures on health outcomes.

It is in this context of the utility and flexibility of GAMs that this approach is planned for the IPD meta-analysis, particularly in the first stage of the analysis. In the case where the generalised linear modelling framework is used, the changes for each type of outcome are based on the choice of link function as described in figure 7 below. In addition, attributable risk of heat to adverse outcomes will be calculated as per IPCC described methodologies.

Outcome type (y_{ij})	Model type	Basic* regression model (for participant j in trial i) to be fitted in each trial separately, including a single covariate (x_{ij} = 1 for treatment group or x_{ij}=0 for control group)
Continuous	Linear regression	$y_{ij} \sim N\left(\mu_{ij}, \sigma_i^2\right)$ $\mu_{ij} = \alpha_i + \theta_i x_{ij}$ NB σ_i^2 specifies that the residual variance is common to the control and treatment groups in trial i; alternatively it could be stratified by trial and treatment group by specifying σ_{ij}^2
Binary	Logistic regression	$y_{ij} \sim$ Bernoulli(p_{ij}) $\text{logit}(p_{ij}) = \alpha_i + \theta_i x_{ij}$
Ordinal (with $C = 1$ to k ordered categories, from worst (1) to best (k))	Logistic regression (proportional odds model)	$y_{ijC} \sim \text{multinomial}\left(p_{ijC}\right)$ where $p_{ijk} = 1 - \sum_{C=1}^{k-1} p_{ijC}$ $\text{logit}(Q_{ijC}) = \alpha_{iC} + \theta_i x_{ij}$ for $C = 1$ to $(k-1)$ where $Q_{ijC} = p_{ij1} + \ldots + p_{ijC}$ and $Q_{ijk} = 1$ and $\alpha_{i1} \leq \alpha_{i2} \leq \ldots \leq \alpha_{i(k-1)}$ NB assumes proportional odds, such that the treatment effect is common (the same) for each category 1 to $(k-1)$
Multinomial (with $C = 1$ to k categories, and category k chosen as the reference)	Logistic regression	$y_{ijC} \sim \text{multinomial}\left(p_{ijC}\right)$ where $p_{ijk} = 1 - \sum_{C=1}^{k-1} p_{ijC}$ $\ln\left(\frac{p_{ijC}}{p_{ijk}}\right) = \alpha_{iC} + \theta_{iC} x_{ij}$ for $C = 1$ to $(k-1)$
Count	Poisson regression	$y_{ij} \sim$ Poisson(μ_{ij}) $\ln(\mu_{ij}) = \alpha_i + \theta_i x_{ij}$
Incidence rate	Poisson regression	$y_{ij} \sim$ Poisson(μ_{ij}) $\ln(\mu_{ij}) = \alpha_i + \theta_i x_{ij} + \ln(t_{ij})$ where $\ln(t_{ij})$ is an offset term for the log of the observed time (t_{ij}) at risk (i.e. the log of the follow-up time)
Time-to-event	Cox regression (proportional hazards model)	y_{ij} distribution unspecified $h_{ij}(t) = h_{0i}(t) \exp(\theta_i x_{ij})$ NB assumes proportional hazards, such that the treatment effect is common (the same) at all time points

* Adjustment for pre-defined prognostic factors is also recommended in practice (Section 5.2.4).
Source: Richard Riley.

Figure 7: Basic format of the regression model to be fitted separately within each trial in the first stage of the two-stage IPD-meta-analysis approach, where the aim is to estimate treatment effects (q_i) from parallel-group randomised trials with continuous, binary, count and time to event data where i denotes trial and j denotes participant and (y_{ij}) denotes the outcome value (61)

For the GAMs the general model formulation specifies for each study a model such that

$$g\big(\mathrm{E}(\mathrm{Y})\big) = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_m(x_m)$$

For the various types of outcomes considered, we can similarly assume the exponential family of distributions to explicitly define the link functions being identity, logit and the proportional hazards function in the case of time-to-event outcomes.

The functions f_i may be functions with a specified parametric or non-parametric form. The case of the former is the use of smooth functions such as polynomials or penalised regression splines, whereas in the latter non-parametric function setting form (for example a polynomial, or an un-penalized regression spline of a variable) or may be specified non-parametrically, or semi-parametrically. The covariates x_i may be multivariate with f_i defining a joint multivariate function or geographically or temporal varying terms.

Pooled analysis

Using the statistical methods described above, an analysis of outcomes of interest for each study will be performed and a summary statistic presented to describe the observed intervention effect. The summary statistic will differ depending on the outcome and the analysis method used.

In the second stage, a weighted average of effects of heat on maternal and child outcomes will be calculated where possible. The weighted average will incorporate the size and magnitude of effect from each study. This will be illustrated in a forest plot. Forest plots will present the point estimates and uncertainty range for each stratum as well as the combined estimates where possible. Forest plots depict the inconsistency across strata and the precision of individual strata.

Exploration of variation in effects across studies and sub-groups

Exploration of variation in effects will be done involving stratified analyses within the following strata: study; geographical area (country or region of country) where all studies in that area are pooled into one estimate; climate zone; time period; and income group of country. Data will also be stratified on population characteristics, such as maternal age, socio-economic status, sex, and conditions such as HIV. In these analyses we generate estimates of effectiveness (aggregate data) for each stratum separately and then combine these summary statistics using standard meta-analysis methods, if appropriate.

Risk of bias across in the Individual Participant data analysis studies

The validity of an IPD is influenced by the completeness and representativeness of the data collected. Using the PRISMA-IPD Flow chart (Annex 1), we will report the numbers of studies screened, assessed for eligibility, and included in the systematic review with reasons for exclusions at each stage. We will describe the distribution of included studies and the characteristics of participants for variables like location, age and prevalence of medical comorbidities. We will compare study level variables between the studies that we collected data from, to those that we were unable to obtain individual data. This will allow us to assess whether there were any systematic differences between these two groups of studies. Drawing on this and factors such as overall rate of participation of eligible studies, we will make an overall assessment of the potential risk of bias associated with non-availability of IPD from cohorts or trials that decline to participate.

Software for individual participant data meta-analysis

The preference in this study will be to use open-source statistical packages to enable reproducibility through sharing of code. From the data pre-processing up to the meta-analysis, the R statistical computing program can be used. R can be used in the Jupyter Hub environment, which will be set up through the Data Management and Analysis Core (DMAC), although there is the option to also use RStudio. There are specific packages that will be used including:

- For fitting nonlinear or linear mixed-effects modelling in R, the *nmle* package can be used for the generalised linear or non-linear mixed-effects as a benchmark prior to implementing GAMs
- For the GAMs - the open-source software libraries that implement the GAMS include *gmcv* the mixed GAM computational vehicle and *InterpretML* to implement boosting and bagging GAM approaches. In R, the *mboost* package can be used to implement boosting
- *IPDmada* is an R shiny based web tool for visually diagnosing IPD models (100), which will be beneficial to explore in the context of this project
- An R package *jomo*, has been developed specifically for multiple imputation of IPD meta-analysis (101)

5.4 Additional analyses

Though analysis will adhere to the protocol, analysis will remain flexible. However, it also creates the potential for multiple testing or "data dredging," whereby analysts explore numerous outcomes and subgroups to find those that yield interesting results.

6. Data management plan

The data management plan aims to describe what data will be used in the study, where this data will be obtained, how the data will be transferred, the platforms and people that will manage the data and what security measures will protect personal data. The data management plan for the HE^{2}AT Center is further detailed in Annex 6 where this IPD is referred to as research project one or RP1.

6.1 Data categories

Multiple categories of data will be used across the project broadly divided into four categories.

1. Health related data

Health related data is generally considered to be personal data as it pertains to individuals' medical records, disease history and health events. In some cases, health related data is also considered *personally identifiable data* where individuals can be identified either directly through names, ID numbers, etc. or indirectly deduced from address, GPS locations, or cross referencing with other identifiers. Many countries now have specific laws dictating conditions for sharing and processing of such data and associated protective measures to avoid unauthorized access and potential harm to individuals involved in the studies. In South Africa the relevant legislation is the Protection of Personal Information Act (POPIA 2013). This necessitates processing in compliance with the specific requirements of the POPIA act, outlined in detail in the section 7 below.

Due to above mentioned privacy, ethical and legal requirements, health related data acquisition almost always involves an associated data sharing agreement (DSA) between the data holder and the HE^{2}AT Center (see data management workflow description below).

2. Climate/weather data

These data include both observational based datasets (weather station observations, satellite proxy observations) and processed/gridded observations. Gridded climate data produced from atmospheric re-analysis and climate simulations will form historical gridded climate observations and forecasts. Climate related data will in almost all cases involve accessing open data repositories such as Copernicus Climate Data Store (CDS) or Earth System Grid Federation data systems. Climate related data will be stored on IBM PAIRS data storage and/or CSAG/UCT data storage systems however CSAG/UCT will manage and update the primary data index for climate related data available to the consortium. All climate datasets used are available through open data policies with no restrictions on non-commercial research use. In some cases, there is a requirement to cite the original source. From Year 5, primary storage will transition to WHC-managed cloud infrastructure (e.g., Microsoft Azure), with dual-system operation maintained during the transition period to ensure continuity.

3. Remote sensing data

The focus is on data derived from satellite sensors, mainly optical imagery (e.g. satellite images of urban centers) as well as indicators of physical measures such as land surface temperature, soil moisture estimates, vegetation condition, land use and cover, etc.) for the purposes of estimating environmental quantities and land use/building density etc. Remote sensing data will not be used to identify individuals in any way and does not constitute sensitive or personally identifiable data. Remote sensing related data will in almost all cases involve accessing open data repositories such as

Copernicus Climate Data Store (CDS), Sentinel data systems, etc. Remote sensing related data will be stored on IBM PAIRS data storage and/or CSAG/UCT data storage systems .From Year 5, primary storage will transition to WHC-managed cloud infrastructure (e.g., Microsoft Azure) with enhanced security measures including AES-256 encryption and 24/7 Security Operations Centre (SOC) monitoring.. All remote datasets used are available through open data policies with no restrictions on non-commercial research use. In some cases, there is a requirement to cite the original source.

4. Area/Geospatial socio-economic data

These data represent measures of socio-economic and related conditions such as household economic status, access to services such as water and sanitation, dwelling type, etc. Typical sources include national census data, and more focused household & demographic survey data.

Socio-economic data will be sourced from both open data repositories as well as restricted access repositories (e.g. South African census data, Gauteng-City Region Observatory (GCRO) Quality of Life Surveys). Primary copies will be indexed and stored on CSAG/UCT/WHC data storage but versions may already exist/can be uploaded on IBM PAIRS to enable analysis through PAIRS. South African census data is already available through the UCT DataFirst data repository and GCRO QoS data is available through the GCRO open data platform as well as through direct queries with GCRO.

South African census data is aggregated up to small areas and so does not constitute personally identifiable data. GCRO Quality of Life survey data is likewise aggregated to small areas and does not constitute personally identifiable sensitive data. These data will be relevant to studies from South Africa.

6.2 Data management workflow

Figure 8 below maps out the data processing workflow. Details of each element within this workflow are described in detail below.

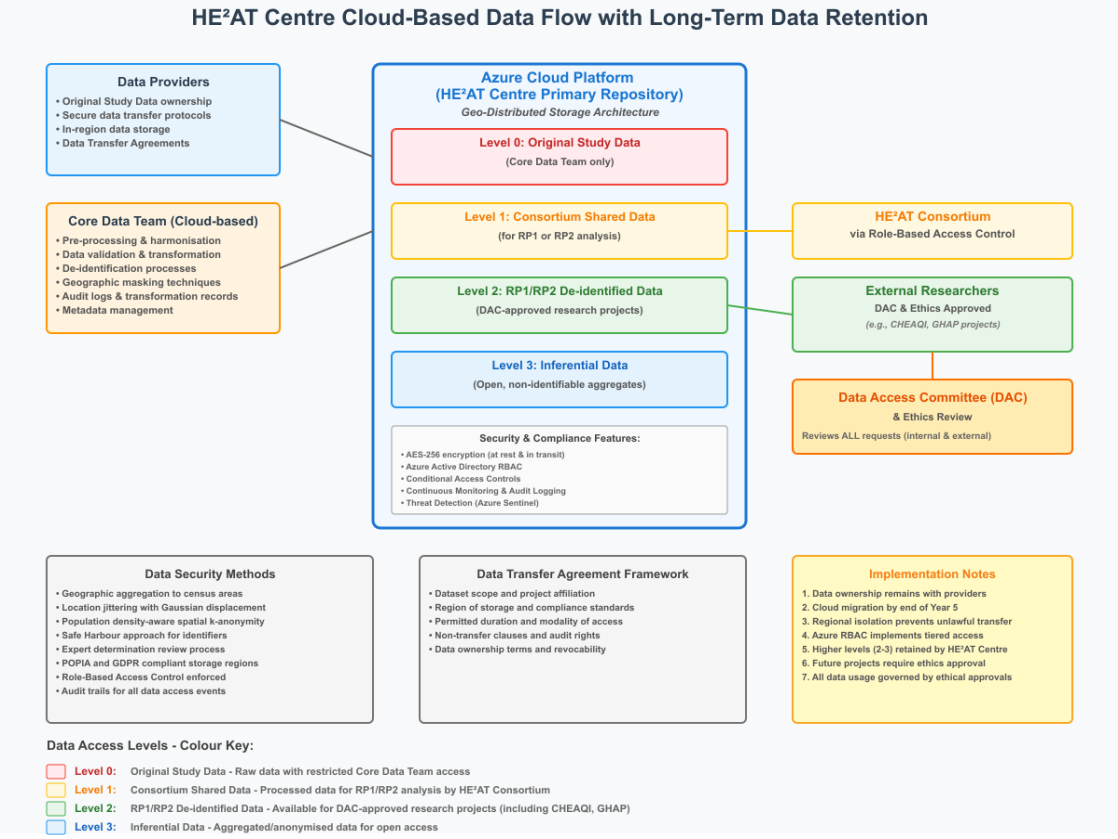


Figure 8: Data flow diagram indicating data providers, data sharing agreements and ethics approval, data harmonization steps, integration, analysis through Jupyter Hub by partners/researchers

Setting up the DSA between the health data provider and CSAG/UCT

As noted above, a DSA will be required between each health data provider and UCT/WHC, in order to enable the transfer of health data to UCT/WHC, and the subsequent processing, analysis, and potentially publishing. Ethical considerations, including informed consent will be addressed, as detailed in section 7, below.

Cloud Infrastructure Transition: From Year 5, the HE2AT Center is implementing a transition from UCT on-premise infrastructure to WHC-managed cloud infrastructure. Data providers will be notified of this transition, which enhances data security with AES-256 encryption and 24/7 monitoring whilst maintaining full POPIA compliance. The transition will be completed by the end of Year 5 through a phased approach ensuring minimal disruption to research activities.

Data encryption and transfer

Once the DSA has been agreed and signed, and no ethical concerns have been raised by the ethics committee, the health data. If the data constitutes personally identifiable data, it will be encrypted by the data provider and transferred through a data transfer service that utilizes encrypted data transport (TLS). For example, the box.com service and many others utilize TLS encrypted transport.

Data storage and encryption

Once transferred to UCT/WHC, if the datasets constitutes personally identifiable data, or if stipulated by the DSA, the dataset will be encrypted for storage using 256-bit AES (Advanced

Encryption Standard) a standard established by the US NIST (National Institute of Standards and Technology) with encryption keys only available to the minimum number of people required to implement any anonymization or data minimalization process. From Year 5, data will be stored on WHC-managed cloud infrastructure with equivalent or enhanced security standards, including automated backup systems and compliance monitoring.

Meta-data (datasets descriptions, descriptions of the original protocols, codebooks, etc.) will be decrypted and stored separately in order to allow for meta-data indexing and software code development.

Data indexing

Data currently available on the CSAG/UCT/WHC data storage system as well as data available on IBM PAIRS will be indexed using an appropriate meta-data standard and this index will be made available by CSAG/UCT/WHC data platform.

CSAG currently implements a **Data Reference Syntax (DRS)** which is a structured mapping from a controlled vocabulary of meta-data elements, to a directory and file naming syntax. This is standard practice within climate data management, and we will continue to implement this approach for the climate and remote sensing datasets. The current DRS is documented on the CSAG/WHC Gitlab Wiki. Integration with DSI-Africa Open Data Science Platform (ODSP) meta-data index will ensure that meta-data propagates to the ODSP system and datasets become discoverable through ODSP meta-data queries. During the transition to cloud infrastructure in Year 5, the DRS and metadata standards will be maintained to ensure continuity and consistency across both on-premise and cloud platforms.

De-identification

De-identification may be implemented (see section on de-identification below) in order to minimize the potential for individuals to be identified either by geospatial information or other personal information.

Codebook remapping and harmonization

Codebook remapping involves translating variables in order to align with a common codebook. We will develop common codebooks defining the required health, socio-economic, other environmental, and climate variables. Each source dataset will require a map that defines how variables are translated from source to the harmonized common dataset. For example, pre-term delivery might be translated from the number of days, to a categorical variable. Clinic names might be translated into administrative area identifiers.

Similar to health data harmonization, climate and environmental datasets will be conformed to a common standard through a separate harmonization process. Environmental data (climate data, remote sensing data, socio-economic mapping data) will be homogenized to align with existing meta-data and storage standards as documented on the Gitlab Wiki. The result of the codebook remapping will enable multiple datasets to be combined into one harmonized dataset including relevant ancillary variables (environmental, climate etc.).

Data integration and analysis

Integrated datasets (with associated documentation provided through the gitlab platform) will be made available for analysis through the Jupyter Hub platform to partner researchers. In some cases,

depending on the conditions of the original DSAs, these partner researchers will have to agree to specific data usage agreements. The full health data processing workflow is mapped in Figure 9 below. The key steps are:

1. Basic error screening and correction to identify formatting errors, invalid codes and values, etc.
2. Screening to check for personal identifying information. For example, date of birth is an indirect identifier, implying high risk of re-identification when combined with other indirect identifiers. Date-of-birth can therefore be dropped, after calculating and keeping the age of the patients.
3. Recoding to one of the identified common codebooks. This step will extract variables of interest, translate variables into the target units including translations of real values into categories, translating from categorisation scheme to another, etc. This step will also include recoding of geographic information. Where point location information is given, the jittering method of perturbing spatial location data will be used, where each point location is off-set in a systematically random way (random distance & random direction). The alternative will be to substitute point locations with grid-cell centroids where the grid-cells correspond to the grid framework to be used for climate & environmental data
4. Further quality control and restructuring of data
5. Integration of climate variables in support of particular analyses

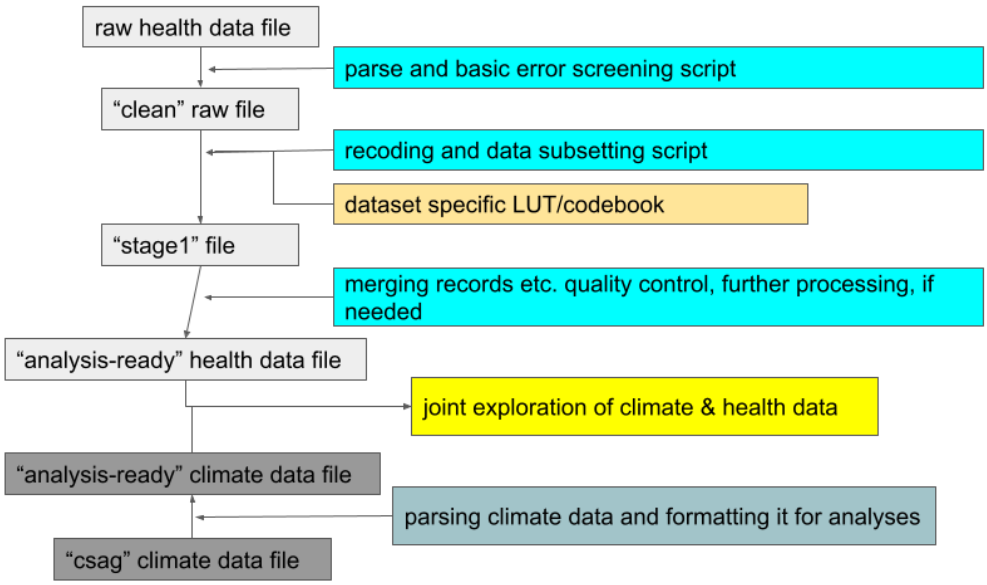


Figure 9: Detailed data processing for health data ingestion

6.3 Data analysis platforms

Data analysis will be enabled through the CSAG/UCT/WHC Jupyter Hub as well as IBM PAIRS:

Juypter Hub is a collaborative web based Python coding environment that enables analysts to develop analysis code in Python using a web based coding platform. The analysis code is executed on the CSAG HPC platform and datasets stored on the CSAG storage servers are available. From Year

5 datasets will be transferred to the WHC-managed cloud infrastructure. The code environment is customized to support all the relevant coding libraries for machine learning (e.g. TensorFlow, PyTorch, Keras, Scikit-learn, Numpy, Scipy, Pandas, etc.).

IBM PAIRS provides multiple interfaces for rapid and more complex analysis which will be made available to HE^{2}AT Center analysts on request (see Roles and Responsibilities for access request contacts). In addition, approaches to integrate data from IBM PAIRS into the CSAG/UCT/WHC Jupyter Hub will be developed.

6.4 Data Management, Documentation and Curation

Data will be managed by the DMAC team and ***primary data management*** will take place on the CSAG/UCT data platform. The data management practices maintaining the IBM PAIRS will continue. Critically, for synergy with the CSAG/UCT platform, there will be sharing of meta-data indexing with CSAG/UCT for data related to NIH HE^{2}AT Center and DS-I Africa. IBM PAIRS will not host health data and so data privacy and access restrictions will be limited to data accessed through the CSAG/UCT platform.

From Year 5, primary data management will transition to WHC-managed cloud infrastructure through a phased migration approach:

- **Phase 1 (Q1-Q2 Year 5):** Planning, assessment, and pilot implementation with limited data migration for testing
- **Phase 2 (Q2-Q3 Year 5):** Gradual migration starting with climate data, followed by progressive migration of health data with enhanced security protocols
- **Phase 3 (Q3-Q4 Year 5):** Complete transition for new data with legacy system integration maintained during transition

Throughout transition: Dual-system operation ensures research continuity with continuous monitoring and adjustment

Individual partner institutions will have their own data and computing platforms ranging from central institutional platforms through to personal computing devices. Data management within these partner institutions does not fall under this data management plan and we refer to this as ***secondary data management***. However, partners will be encouraged to regularly consider what locally managed data should be integrated into the project wide data management platform, and made available to the broader DSI-Africa programme and beyond. It is critical that health data be kept in a secure and managed platform, therefore where researchers from the various partner institution wish to run analyses on the health data in the CSAG/UCT/WHC platform, they will use the JupyterHub environment from which they will also be able to access data from the PAIRS system through an SDK.

Primary data management will involve:

- Homogenization to agreed data archive standards (see above for each class of data)
- Version control of datasets that are regularly updated to ensure prior version remain accessible
- Harmonizing of health related data
- Meta-data indexing within the CSAG/UCT/WHC DMP as well as DS-I Africa ODSP

- Documentation of data on CSAG/UCT/WHC GitLab wiki
- Sharing of data management code through CSAG/UCT/WHC GitLab code repositories

6.5 Data Security and Confidentiality of Potentially Sensitive Information

6.5.1 De-identification

POPIA Section 10 prescribes the principle of Minimality which means that *only information relevant to the purpose of processing should be processed*. Where personal information is acquired that is required to fulfil the research purposes described by the relevant research project protocols, de-identification will be implemented according to the following protocol which is guided by US Department of Human and Health Services (HSS) guidelines.

Street addresses may be aggregated into geographical regions such that many records map to the same region and it is no longer possible to derive individual residential locations. Aggregation will also consider the number of records that map to the same geographical area.

Latitude/longitude coordinates may be "jittered" which involves adding random values to each coordinate such that the exact location is lost (Zandbergen 2014 (89) assesses different approaches to location masking and jittering), while retaining sufficient geographical information to support the analysis. This will likely be the case where jittering will be used to shift latitude/longitude locations in the order of 10s of kilometers, adequate to prevent locating individual residential locations.

6.5.2 Data encryption

Primary data (before processing) will be stored in encrypted form using 256bit AES (Advanced Encryption Standard) a standard established by the US NIST (National Institute of Standards and Technology) with encryption keys only available to the minimum number of people required to implement the de-identification processing. The use of cryptographic modules validated to NIST FIPS 140-2 is required by the United States Government for encryption of all data that has a classification of Sensitive but Unclassified (SBU) or above.

For datasets that include personal identifiers (i.e. personally identifiable data), datasets will be encrypted by the original data holder and transferred to UCT/WHC through a secure data transfer that uses TLS encryption.

6.5.3 Storage isolation

Encrypted data will be stored in an isolated virtual storage server that is only accessible to the private data team.

6.5.4 Network firewall and virtual private network

CSAG compute infrastructure sits within the UCT intranet and falls under the broader UCT security policies and services. This ensures that network access to CSAG servers and computer services is carefully controlled and limited to authorized users only.

UCT broader security policy

http://www.icts.uct.ac.za/sites/default/files/image_tool/images/286/UCT_Information_Security_Policy_PC03_2020.pdf

UCT ICTS implements a Cisco firewall and Cisco Virtual Private Network service to ensure that intranet access is limited to authorized UCT users only.

The UCT firewall policy describes the UCT firewall implementation which ensures that only authorized access to the UCT intranet occurs.
http://www.icts.uct.ac.za/uct-perimeter-firewall-policy

These network level security measures ensure that access to the UCT intranet is controlled to authorized UCT users. Furthermore, CSAG server access is controlled by CSAG authentication and authorization as described below.

From year 5, the WHC-managed cloud infrastructure will implement equivalent security standards.

WHC database security and cloud policies hyperlinks:

- Information security policy and privacy standards
- Data Access Management Policy
- Information Classification Policy
- Firewalls policy
- Security Standards: Applications, Databases, Network, Operating Systems

The WHC-managed cloud infrastructure implements:

- Cloud-native firewall services with equivalent protection to UCT's Cisco firewall
- 24/7 Security Operations Centre (SOC) monitoring and advanced threat detection
- Role-based access controls with multi-factor authentication
- Automated compliance reporting for streamlined governance oversight

6.5.5 Local authentication and authorization

While CSAG compute servers sit within UCTs intranet, the CSAG/UCT platform (storage and compute infrastructure) implements an independent authentication and authorization service (Linux filesystem and LDAP authentication). Access to restricted datasets will therefore be ensured through both UCT authentication protocols and internal CSAG data management plan (DMP) authentication and authorizations.

From the start of Year 5 and during the transition period, enhanced cloud security measures include:

• AES-256 Encryption: All data at rest and in transit

• 24/7 SOC Monitoring: Continuous security monitoring and threat detection

• Role-based Access Controls: Multi-factor authentication and granular permissions

• Automated Backup Systems: Enhanced disaster recovery capabilities

• Compliance Monitoring: Automated reporting for regulatory compliance

6.6 Data sharing and open access

According to the Research Data Management policy, 'publicly funded research data are a public good, produced in the public interest, which should be made openly available with as few restrictions as possible in a timely and responsible manner'. Data is therefore open by default, closed by exception (e.g. privately-funded research, or research with commercialisation possibilities).

Restrictions to data sharing

According to Section 4.6 of the UCT Research Data Management policy, "[n]ecessary constraints on the availability of data include the protection of personal data; the protection of intellectual property; the protection of commercial interests of project partners; and security concerns."

Strategies to limit restrictions may include data being de-identified (see above), gaining participant consent for data sharing; and gaining copyright permissions.

Discoverability

The HE^{2}AT Center and DMAC will implement FAIR principles:

Data will be **findable** through publicly accessible and searchable meta-data indexes (need to decide if the DS-I Africa ODSP is going to be the primary mechanisms for meta-data search, UCT also hosts the ZivaHub repository)

Data will be **accessible** either openly through a public facing component of the DMP data repository, or through a data access request to DMAC (where a Data Sharing Agreement is required)

Interoperability will be enabled through the strict adherence to established data and meta-data standards (see above)

Reuse will be supported through rigorous documentation of the data including limitations and guidance for reuse.

Procedure for making data available to qualified individuals

The HE^{2}AT Center Steering Committee (SC) will develop a Data Access Request Form which people requesting data need to complete prior to consideration of their request. The form will include a proposal outline of the intended research, and the procedures to maintain data confidentiality and security.

The SC will review and approve or reject all requests from the research community, which includes scientists or medical professionals working at academic, non-profit or government institutions, or commercial companies. We will ensure that all requests conform with NIH policies and procedures including compliance with informed consent procedures if relevant, and any limitations stipulated by the institutions/investigators who contributed data. The study will comply with the principles of the Data Protection Act of the country of the participating site. Some countries may have restrictions on data sharing outside of the host country, which we will abide by. Researchers who request to share the resources of the HE^{2}AT Center will need to agree to not seek to identify individuals within the dataset, not distribute the data to any other entity, keep the data secure, and acknowledge the HE^{2}AT Center and DS-I Africa as appropriate in publications and presentations (the exact

acknowledgement text to be agreed among the DS-I Africa Program). Further, researchers who share the Center's resources will be strongly encouraged to collaborate with and train African investigators as part of the work they conduct with the resources shared.

There may be exceptions to the resource sharing plans outlined above. Firstly, there may be considerations around intellectual property protections for the research products that the consortium aims to commercialize. Decisions about resource sharing in these circumstances will follow the NIH policies in this regard, including those on resource sharing, disseminating unique research resources, and program income. The NIH will be provided with a copy of documents or samples of these products developed under the grant award (e.g. the Digital App). A large portion of the data we will use in the Project activities will be drawn from the IBM-PAIRS platform, which contains several datasets that require a license for access. We will thus not be able to share those datasets without permission from the licensee.

Clearly, the rights and privacy of individuals who participate in research must be protected at all times. Thus, data intended for broader use should be free of identifiers that would permit linkages to individual research participants and variables that could lead to deductive disclosure of the identity of individual subjects. All data shared beyond the HE^{2}AT Center will be de-identified following the procedures described above. Requests for original data containing personally identifiable or sensitive information will be referred back to the original study.

The HE^{2}AT Center team will especially endeavour to make the unique research resources that we develop readily available to other researchers working on climate and health. There is a pressing need within the field to expedite the translation of data into knowledge and, in particular, into interventions that protect people against extreme heat and other manifestations of climate change. We thus feel obligated to expedite our responses to such data requests. Fellow researchers wishing to access the data will thus not have to wait for our research findings to be accepted for publication or for the 'final research data" prior to data sharing, provided the research questions they are exploring do not directly overlap with the specific question we are addressing.

The IPD involves 'data sharing' on a large scale, where we are reliant on the willingness and ability of data providers to share. In that spirit, as the holders of the IPD database, it will be beholden on us to share the data with other groups, provided the original data holders agree to sharing and the procedures of HE^{2}AT Center data sharing processes are followed. We will develop formal collaboration agreements around data sharing with the data providers who contribute data to the IPD, which set out the terms and conditions for data reuse. This is important, as for example, some of these investigators may not wish their data to be shared with third parties, for example. While a willingness to share data forms the basis for an IPD and most often reflects a desire of investigators to collaborate, this may not necessarily translate into a willingness to then share data beyond the Center. Sharing of the datasets gathered into the IPD database will thus require signed agreement from the original data providers and may require an application to the local IRB where the original study was done. People who make use of the IPD database will have to undertake to adhere to the authorship guidelines stipulated in the collaboration agreement signed between the HE^{2}AT Center and the research groups who contributed their data.

Of note, much of the data and resources generated by the Hub will be useful for a range of other disciplines. The datasets we will generate in the IPD have tremendous potential to answer a wide range of questions, outside of environmental health. We will facilitate access to these resources by interested parties. We also undertake to share data and resources generated by the HE^{2}AT Center

with other Hubs or components in the DS-I Africa program, wherever possible. In particular, the DS-I Africa Open Data Science Platform will be used as a mechanism for making relevant datasets from the HE^{2}AT Center available across the program, and beyond.

In terms of the DMAC activities, data and data resources (primarily software/code) will be shared through standard protocols such as File Transfer Protocol (FTP) servers and the UCT/WHC instance of Gitlab (software version control system). From Year 5, cloud-based data sharing protocols will be implemented whilst maintaining the same security and access control standards. UCT/WHC also hosts an open data portal which will be used to make more final datasets available following FAIR principles. Data that is made publicly or otherwise available through data use and sharing agreements will be accompanied by metadata such as data dictionaries and data descriptors such as principal investigator, funding sources, data collector, project description, sample and sampling procedures, temporal and geographic coverage of the data collection, variables, technical information on files (file formats, linking, etc.), interviewer guides and coding instruments.

Analytic data sets will be provided as de-identified data files that can be read by common statistical package software, such as SAS or Stata. De-identified data sets will have names and other personal health identifiers removed (see above). Data and other resources will be transferred to others under the terms of a data sharing agreement to ensure that the data will be used for the proposed purpose and that no attempts will be made to identify participants. We will maintain records of all researchers who have been given access to the research resources. Results of the HE^{2}AT Center activities will be shared with the research community and the public through conference presentations, publication in peer-reviewed journals and media interactions, as described in the publications and dissemination below in section 9.

6.7 Data retention

Participant data and the data of the children will be retained for a period for at least 5 years after the completion of the project for historical, statistical or research purposes as provided for by POPIA. Appropriate safeguards will be established to secure the records and ensure that the data is not used for any other purpose than the purpose for which the data was originally collected. In any event, no records will be retained for longer than as per applicable regulations.

7. Ethical considerations and legal requirements

7.1 Summary of general ethical and legal requirements

The project raises several ethical and legal considerations. This includes 1) the use of secondary data from pregnant women and children for research purposes that may differ from the original purpose of the study, 2) sharing data across countries in Africa, and 3) the presence of potential identifying information.

The project will endeavour to ensure the protection of human subjects and their data through procedures that adhere to the following legal and ethical guidelines:

- Declaration of Helsinki
- ICH Good Clinical Practice
- Ethics in Health Research (Department of Health South Africa) (102)
- Protection of Personal Information Act (POPIA)
- Any additional African country-specific protection of data legislation

Table 7 below summarises some of the ethical and legal considerations for the project, and how these will be addressed. Additional details describing these procedures are described thereafter and further detail included in the Data Management Plan in Annex 6.

Table 5: Summary of ethical, legal and data management considerations that underpin study activities

Ethical considerations	Legal requirements	Data management activities
• The purposes of the research are clearly defined in the IPD protocol and the reasons that the study needs to involve personal data are provided • Processing of historical data for research purposes in the IPD serves the public interest • Specific concerns of vulnerable populations such as pregnant women, neonates, and children have been considered • We assess ethics concerns in the studies that contribute data, including information on the ethics committee that provided approval for the original study, securing a copy of the informed consent of the study, if possible, and clarifying any concerns about the study with the data provider, where feasible • Participant informed consent has been addressed, separated into two scenarios: 1. Participant has signed a ''broad consent'' for use of their data in future research projects, even though the precise nature of the future research is unclear, this will allow data sharing without further ethical approvals 2. Participants that have signed ''narrow consent'', where sharing of data is not permitted, and/or "tiered consent", where each individual participant chooses to	• POPIA requirements are met • The data provider has determined that the country's legal requirements for personal data security are adhered to • Data sharing agreements are in place for each dataset from data provider to UCT • Data sharing agreements specify the use of the data through UCT's database by other partners like IBM, CeSSHAR and Wits RHI • Any time limitations on use of data are adhered to	• A data management plan clearly defines how the data is transferred, stored and secured • Clearly defined and limited personnel will have access to data through access-controlled platforms, with specific individuals authorized for different levels of data access (e.g. only specific named individuals have access to encryption codes, and a log of all those who have access to the data for analysis) • Only data that is necessary for the conduct of the study will be collected and data will be anonymised, when required • Publications will not include any potentially identifiable information from participants, and only aggregated data will be published or presented publicly

permit data sharing or not, will be carefully considered: ▪ The possibility for reconsenting will be deliberated if no consent for data sharing has been provided by the participant. In cases where this is feasible (e.g. ongoing studies), reconsenting is done ▪ If reconsenting is not feasible, impossible, or would involve a disproportionate effort, an informed consent waiver will be requested from the UCT Faculty of Science Research Ethics Committee • Ethics committee has approved the conduct of the IPD study and are provided with information on each of the studies that contribute data • Risks to individuals' privacy and potential for identification are limited as far as possible		

7.2 Assessment of the informed consent process in primary study

The informed consent process for each dataset that agrees to contribute data to the IPD will be assessed for ethical considerations. Firstly, we will review informed consent procedures that were undertaken by the participant for the primary study. If a participant has signed a ''broad consent'' for use of their data in future research projects, even though the precise nature of the future research is unclear, this will allow data sharing without further ethical approvals.

Participants that have signed ''narrow consent'', where sharing of data is not permitted, and/or "tiered consent", where each individual participant chooses to permit data sharing or not, will be carefully considered. The possibility for reconsenting will be deliberated if no consent for data sharing (narrow and tiered with no consent) has been provided by the participant. If reconsenting is not feasible, impossible, or would involve a disproportionate effort, an informed consent waiver will be requested from the Wits Human Research Ethics Committee.

7.2.1 Waivers of consent for use of data for purposes other than that stipulated in the informed consent form

Waivers of informed consent are relatively common in public health settings. For example, in instances where data are used on individuals or groups to prevent the spread of communicable diseases, as occurred in the COVID-19 pandemic. Oftentimes health information is collected without consent to determine the incidence of various health conditions. This is analogous to our study where we wish to calculate the incidence of adverse birth outcomes during hot periods, compared to cold periods. The justification for these breaches of the normal right of individuals to informed consent is that this right can be subordinated to the needs of the community if the two are in conflict. Naturally, the extent to which this subordination applies to research needs to be determined for each research proposal.

The informed consent requirements for participation in research are stricter than those for both individual and public health interventions because the research participants are exposed to the risks of research but often do not receive any benefits even when the research is successful. In contrast,

public health interventions, such as programmes to reduce the impacts of extreme heat on health, often do benefit the recipients of the intervention. In the absence of a waiver, the research would be severely impeded, or not possible in fact.

We acknowledge that for research using identifiable human data, researchers must normally seek consent for the collection, analysis, storage and/or reuse. The Declaration of Helsinki, paragraph 25 notes that it is preferable that potential research participants be informed in advance about any possible future uses of their data so that it will not be necessary to re-contact them when another such use is proposed. When the information in databases are anonymized and aggregated, it is often impossible to identify the source persons from whom to ask informed consent for new uses. This is especially true for databases that were established in the past when informed consent for future uses was not sought. Research ethics committees may decide to waive the informed consent requirement by considering the Declaration of Helsinki's exception to the requirement as stated in paragraph 25: “There may be situations where consent would be impossible or impractical to obtain for such research or would pose a threat to the validity of the research. In such situations the research may be done only after consideration and approval by a research ethics committee.” The CIOMS International Ethical Guidelines for Epidemiological Studies (commentary on Guideline #4) also provides support on this exception.

7.3 POPIA compliance and protection of personal information

The use of health datasets requires careful consideration of data security and confidentiality. This is strongly guided by the relevant legislation context for each datasets, including the specific country legislation around the use of personal/sensitive data, and the cross border transfer of such datasets.

The Protection of Personal Information Act of South Africa (2013) institutes limitations on the processing of personal information, but also provides the legal basis for the use of personal information for the purpose of scientific research. POPIA functions alongside other legislation and regulatory structures governing research in South Africa, such as the Constitution of the Republic of South Africa, the National Health Act No 61 of 2003 and regulations as prescribed by the Minister of Health, such as the South African Department of Health guidelines on Ethics in Health Research Principles, Processes and Structures 2015 (“DOH Guidelines”). The law which takes precedent will be that which provides the most comprehensive protections to the rights of individuals in South Africa.

Section 6 of the Act indicates that the Act does not apply in cases where personal information has been de-identified to the extent that it cannot be re-identified again. Where this is the case, the information can be used without restriction or conditions of the Act. It is envisaged that a number of health databases acquired for the project will have been de-identified to the extent that re-identification is virtually impossible. However, where this is not the case, the following sections of POPIA provide a basis for processing health data.

While Section 15(1) requires that personal information processing must be compatible with the purpose of collection, Section 15(3)(e) allows for processing for the purposes of historical, statistical and research purposes regardless of the original purpose of collection. As the IPD will be using health datasets collected prior to the project, we will never be processing personal information compatible with the original purpose of collection, unless specific consent for further processing was agreed to by the data subjects. Section 15(3)(e) therefore provides the basis for processing information where this specific consent was not obtained.

Furthermore, while Section 18(1) lays out requirements for informing the data subject that their personal information is being processed for a particular purpose, Section 18(4)(f) provides for an exemption from these requirements where the information will be used for historical, statistical or research purposes.

Furthermore, Section 14(2) allows for the retention of personal information for research purposes as long as safeguards are in place to prevent the use of the personal information for other purposes. Section 19 requires that security measures are implemented to prevent the unlawful access to or processing of personal information. We will implement extensive data security measures. Section 20 describes requirements around individuals operating on or processing personal information. We have identified the individuals who will be responsible for this processing and will continually update this list. Access to personal information is restricted through passwords and other security measures so that only individuals authorized by the responsible party have access.

Section 21 requires a written contract between the responsible party and the operators implementing processing. This contract will be signed by all individuals processing personal information and is presented in the data management plan and includes agreement that the operator will inform the responsible party where there is any basis for believing that personal information has been accessed by an unauthorized person.

As noted in Section 27 of the POPIA Act, in instances where the study investigators are requesting access to data for a purpose which was not originally agreed upon by the participant in the informed consent, we will only use the data for research and statistical purposes.

We believe that the inclusion of such data for the research purposes set out in this protocol is justified as it serves a major public interest, and the processing of data is necessary for the purposes concerned. Climate change, and in particular, rapidly escalating temperatures and heat waves, are a major threat to health across the world, and especially in Africa. Many parts of Africa are warming at twice the average warming rate. Additionally, there is little research on the impacts of extreme heat on pregnant women and children in Africa and this information is vital for developing services to assist these population groups as temperatures escalate. There is thus a major public interest to understand the implications of high temperatures on health in Africa. This information will help to inform the allocation of funding to ameliorate these impacts and assist in selecting appropriate interventions in this regard.

7.4 Data sharing compliance

The data provider will be required to provide a contractual assurance in the data sharing agreement that informed consent procedures were followed for the original studies and that they have individual participant consent to share the data for further scientific research of this nature, in compliance with all data protection legislation applying in the host country.

7.5 Considerations specific to data source

As described in section 4.6, data will be sought from various sources, such as data repositories, from research sites run by Wits University, from other universities or research organisations in South Africa and across sub-Saharan Africa. The ethical and legal considerations for each study may vary by data source, as presented in Table 8 below.

Table 6: Ethical and legal considerations for different types of data

Types of data sources			Approach	
Type	**Description**	**Example**	**Ethical**	**Legal**
Open source	Available online for download, no data sharing agreements required	Reassessing pregnancy intention and its relation to maternal, perinatal and neonatal outcomes in a low-income setting: A cohort study. https://rdr.ucl.ac.uk/articles/dataset/Pregnancy_intention_and_perinatal_outcomes/8949050	- No further ethics submissions or approvals required - Data will be treated with equal protection standards as other datasets	- Assess study individually for any terms and conditions of sharing (may have embargos on duration of access etc.) - Data transfer is subject to country specific legal requirements
In data repository	Access restricted and study staff need to request data, which is reviewed prior to approval	Worldwide Antimalarial Resistance Network	- The data provider will be responsible for their own ethics committee submission, if required	- Complete data requests forms as required by data repository - Complete DSA if required - Adhere to terms and conditions of sharing (e.g., time embargo) - Data transfer is subject to country specific legal requirements
Wits University (including Wits Health Consortium)	Participants have signed broad informed consent (and possible IPD sharing statement)	ADVANCE study	- No further ethics submissions or approvals required - Participants have agreed to sharing data - Ethics committee to be informed about the study contributing data and provided with related documentation	- DSA to be completed - POPIA compliance
South African university/organisation (other than Wits)		SAFE Passage – large cohort from Stellenbosch University	- Assess if participants have signed broad consent - Explore feasibility of reconsenting participants - If reconsenting not possible, apply for informed consent waiver to the UCT Faculty of Science Research Ethics Committee	- DSA to be completed - POPIA compliance

Study and/or PI is from another country	Study was conducted in an African country that is not South Africa – can be single- or multisite study	Kesho Bora study: Maternal ART during pregnancy and breastfeeding prevents more infections than short-course prophylaxis. Study conducted in Kenya, Burkina Faso, and South Africa	- Data provider will be responsible for ethics submission to their own ethics department, if required - Explore informed consent processes	- DSA - POPIA compliance - Compliance with personal data legislation in other countries (data provider is responsible), including cross-border data transfer laws

The Data Management Plan (Annex 6) provides additional detail on the steps in which ethical and legal standards will be upheld in the acquisition, handling, and safekeeping of data. As the data will be transferred to a server at UCT, the Faculty of Science Research Ethics Committee will be approached for any additional study specific ethics approvals, if required. As additional databases become available to the study team and DSAs are agreed to, we will notify the UCT Science Faculty Ethics Committee in writing of the new study that has agreed to contribute data to the study. We will provide information about the database prior to actual data transfer. We will inform the committee of the name of the study, study acronym, contact details of the data providers, details of the ethics approval of the original study including informed consent parameters, where available, study country and provide a copy of the DSA. We will make additional information available on the study where requested by the ethics committee. The University of the Witwatersrand Human Research Ethics Committee will be provided with documentation of the studies that have been included in the IPD and documentation of the decisions made by the UCT Ethics Committee in this regard, six-monthly, or sooner where requested.

Risks to the participants in the primary studies

Risks to a participant in this study are minimal as there are no new procedures involved and steps will be taken to protect their personal information and to avert risks of identification, as described in the data management plan.

8. Training, engagement, and capacity building

Coordinated and consistent implementation of health system operations can be effective and cost-efficient but are labour and resource intensive. For health adaptation interventions to decrease the number of heat-health risks among vulnerable populations, there needs to be a parallel effort to ensure response capabilities are developed that take advantage of improved surveillance systems. Working with the ministries of health to build a response system that is useful, intuitive, and capable of being implemented for long-term use is critical. For example, high-risk maps can be translated and shared with partner countries to be integrated into appropriate communication and adaptation planning mechanisms.

Through the Training and Engagement Core (TEC), the HE²AT Center will serve as a Center of Excellence and resource for the DS-I Africa Research Training Program. The HE²AT Center will engage with the communities in which the research is conducted, to increase acceptability of the study and its objectives. These activities use existing engagement mechanisms, engaging with local Community Advisory Boards and media and distributing newsletters. Because the solutions developed by the Center address two major concerns for policymakers (how to warn people about a heatwave and then to track its impacts), we anticipate that local, national and international policymakers will be willing to engage.

The specific aims of the TEC are to:

- Advance knowledge and skills of early career researchers and scientists of the HE²AT Centre and other Research Hubs on transdisciplinary heat-health research using data science approaches
- Build knowledge, skills, and competencies in solutions for addressing heat-health impacts, particularly early warning and response systems
- Develop and implement a systematic strategy for forming, maintaining, and strengthening engagement with governments, industry, non-governmental and related organizations, and communities
- Disseminate results through scientific publications and presentations at scientific meetings, community engagement, policy engagement, and reports to NIH.

Approach

The TEC engagement plan will take a phased approach to prioritize different stakeholder groups at different stages of the program lifecycle. Given some of the potential research outputs, it will be important to think about engagement across the implementation phases (e.g., pre- mid – post) – with a focus on the pre-implementation stage in Year 1 and 2, with mid-level implementation starting in Year 3. Across the IPD study, there are several points of engagement needed, ranging from global to local levels (Figure 4). The TEC will aim to address needs across multiple scales during the lifespan of the study.

Utilizing the working group for policy and stakeholder engagement the TEC will provide targeted support to the team to help meet the specific needs of end-users, including linking research outputs to policy interventions and building awareness through the documentation of health impacts of extreme heat. A secondary goal of the engagement working group will be to foster learning opportunities across HE²AT Centre teams to strengthen knowledge engagement in the context of climate change and health. This may include inclusion of presentations to the technical talk series

focusing on engagement approaches, as well as journal clubs within/across working groups to discuss relevant literature.

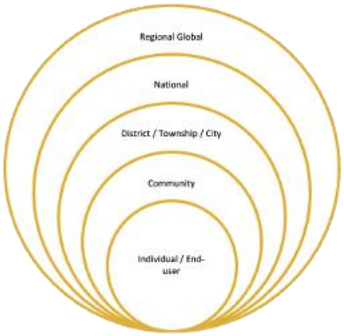


Figure 10: Levels of stakeholder engagement

The TEC and engagement working group will provide technical support for the following activities:

- Conducting a stakeholder mapping exercise and workshop to identify and assess needs of decision-makers and practitioners, particularly related to district level surveillance systems and community advisory boards.
- Conducting a heat-health policy review and analysis for selected study locations
- Conducting a technical needs assessment and training for health workers to build awareness of heat-health risks, as well as identify, prioritize, and implement intervention strategies.
- Supporting the engagement with communities and decision-makers to describe the study goals and gather inputs on the findings and potential operationalization of intervention strategies.
- Conducting capacity building exercises with key stakeholders including end-users and decision-makers to ensure core competencies are established and knowledge increased to utilize study information and implement response actions.

9. Publications and dissemination

The HE^{2}AT Center will disseminate study findings and other learnings in a timely manner across our consortium, partners, policymakers, the scientific community, other stakeholders, and the general public to maximise its impact. Our engagement plan includes:

- Scientific publications and presentations to scientific conferences
- Community engagement
- Policy engagement

Scientific publications will be published in open access journals with data sharing. The final protocol for this project will be made available on open access to allow replication of the approach in other settings and will be registered with PROSPERO.

Each year there will be presentations to scientific conferences, such as the International Society for Environmental Epidemiology, the NIH biannual meetings, the EDCTP Forum, and others. Presentations will also be made upon request to meetings organised by the African Regional Office of the World Health Organization, the United Nations Environment Programme, the United Nations Development Programme, and others. Further, project findings and communication materials may be shared via the Global Heat Health Information Network. The emphasis will be on promoting junior researchers to increase their confidence and skill in preparing and delivering scientific presentations.

The HE^{2}AT Center will promote the project and its findings among the population and the communities in which the research will be conducted, to ensure increased acceptability of the project and their objectives, as well as promote awareness of heat-health risks. The communities and populations in which this research is being conducted are vast; we will engage with stakeholders in maternal and child health in Africa to promote awareness and use the project outputs to outline high risk populations, where engagement can then be focused. Dissemination tools such as newsletters, brochures, introductory project posters, and community advisory board discussions will be utilised to engage with various communities. We will emphasise the guiding principles of Good Participatory Practice, including respect, mutual understanding, integrity, transparency, accountability, and autonomy. Some of the team members already engage frequently with the media about climate or extreme heat concerns, which opens further avenues for the dissemination of findings.

Project results will be disseminated to *local, provincial, and national authorities*, to engage with, provide technical support, and inform policies, for example, Climate Change and Health Adaptation Planning. The Africa CDC will also play a major role in dissemination activities of the HE^{2}AT Center and are fully committed to participate actively in the activities of the Center, including hosting joint dissemination meetings and/or stakeholder consultations. In addition, some investigators on the HE^{2}AT Center are active members of the CHANCE network (Climate-Health Africa Network for Collaboration and Engagement) which was established to facilitate the creation of a channel of communication and to establish coherence for various stakeholders in climate change and health in Africa. The network is open to national policy makers, climate services partners, humanitarian organizations, NGO's, research institutes, individual researchers and other stakeholders based in Africa and can be utilised for engagement on the HE^{2}AT Center.

10. Milestones and timelines

The Gannt chart (Figure 11) represents the expected milestones and timelines for the project. The start date is September 2021.

	Sep 21–Jun 22			July 22-Jun 23				July 23-Jun 24				Jul 24-Jun 25				Jul 25-Jun 26				CO* (Jul 26-Jun27)
Milestones **Project quarters**	**1**	**2**	**3**	**1**	**2**	**3**	**4**	**1**	**2**	**3**	**4**	**1**	**2**	**3**	**4**	**1**	**2**	**3**	**4**	**1-4**
Initial IRB and ethics approval																				
Training of HE²AT Center team on RP1 methods																				
Stakeholder engagements																				
Systematic search of eligible studies in duplicate and extraction of data																				
Completion first three sections of PRISMA IPD flow diagram (identification, screening, eligibility)																				
Contacting data holders																				
Completion of PRISMA diagram																				
Two-stage statistical analysis of acquired data																				
Biomedical data acquired, recoded, cleaned and integrated into the IPD platform & integration with environmental data																				
Machine learning approaches to understand and quantify the impact of heat exposure on pregnant women, neonates and children																				
Manuscript writing																				
Dissemination meetings and conferences																				

Figure 11: Gantt chart representing the timelines and milestones

*Close-out

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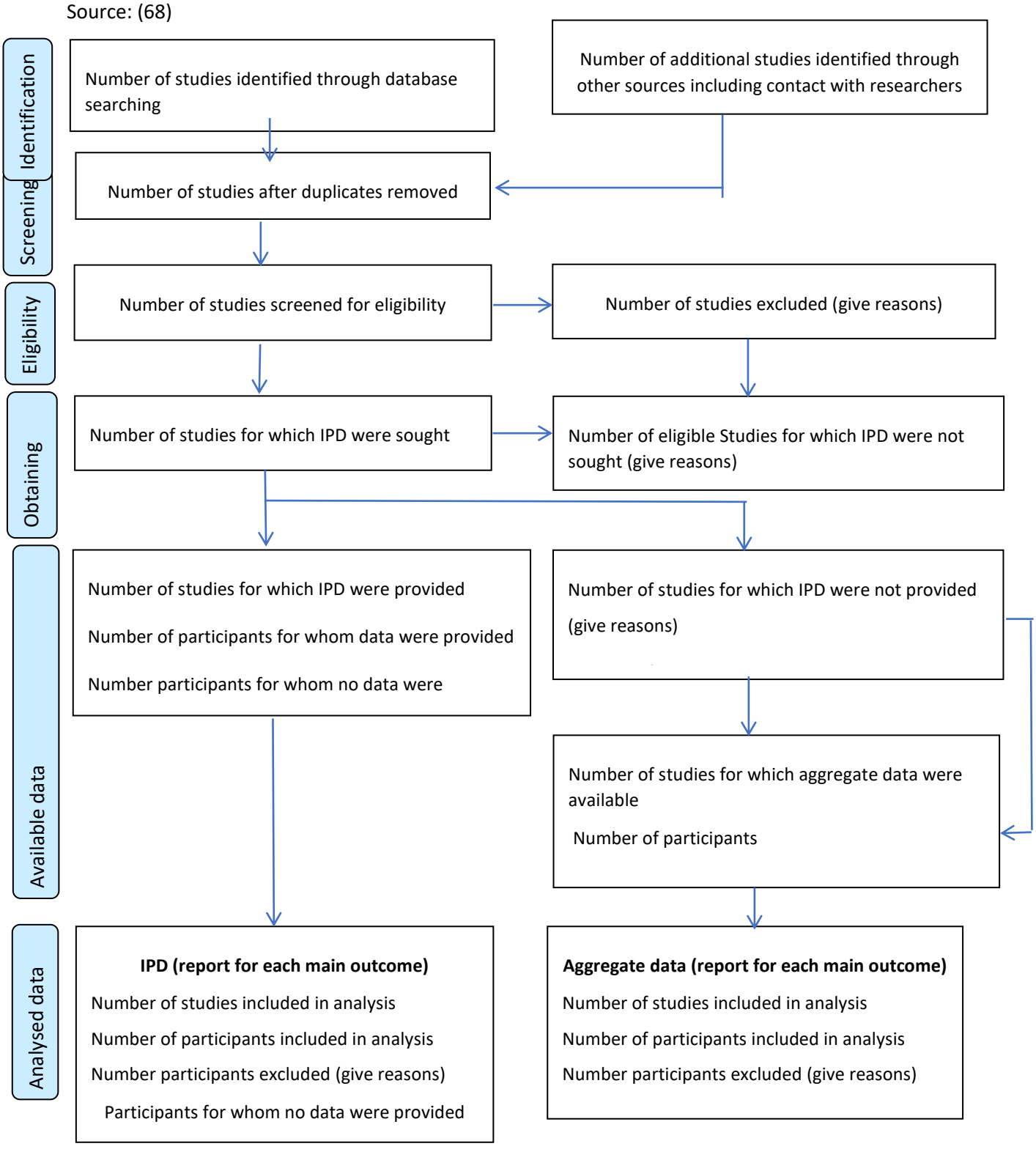
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12. Annex 1: PRISMA IPD Flow Diagram

Source: (68)



The PRISMA IPD flow diagram

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13. Annex 2: PRISMA-IPD Checklist

PRISMA-IPD Checklist of items to include when reporting a systematic review and meta-analysis of individual participant data (IPD) (68)

PRISMA-IPD Section/topic	Item No	Checklist item	Reported on page
Title			
Title	1	Identify the report as a systematic review and meta-analysis of individual participant data.	
Abstract			
Structured summary	2	Provide a structured summary including as applicable:	
		Background: state research question and main objectives, with information on participants, interventions, comparators and outcomes.	
		Methods: report eligibility criteria; data sources including dates of last bibliographic search or elicitation, noting that IPD were sought; methods of assessing risk of bias.	
		Results: provide number and type of studies and participants identified and number (%) obtained; summary effect estimates for main outcomes (benefits and harms) with confidence intervals and measures of statistical heterogeneity. Describe the direction and size of summary effects in terms meaningful to those who would put findings into practice.	
		Discussion: state main strengths and limitations of the evidence, general interpretation of the results and any important implications.	
		Other: report primary funding source, registration number and registry name for the systematic review and IPD meta-analysis.	
Introduction			
Rationale	3	Describe the rationale for the review in the context of what is already known.	
Objectives	4	Provide an explicit statement of the questions being addressed with reference, as applicable, to participants, interventions, comparisons, outcomes and study design (PICOS). Include any hypotheses that relate to particular types of participant-level subgroups.	
Methods			
Protocol and registration	5	Indicate if a protocol exists and where it can be accessed. If available, provide registration information including registration number and registry name. Provide publication details, if applicable.	
Eligibility criteria	6	Specify inclusion and exclusion criteria including those relating to participants, interventions, comparisons, outcomes, study design and characteristics (e.g. years when conducted, required minimum follow-up). Note whether these were applied at the study or individual level	

		i.e. whether eligible participants were included (and ineligible participants excluded) from a study that included a wider population than specified by the review inclusion criteria. The rationale for criteria should be stated.	
Identifying studies - information sources	7	Describe all methods of identifying published and unpublished studies including, as applicable: which bibliographic databases were searched with dates of coverage; details of any hand searching including of conference proceedings; use of study registers and agency or company databases; contact with the original research team and experts in the field; open adverts and surveys. Give the date of last search or elicitation.	
Identifying studies - search	8	Present the full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	
Study selection processes	9	State the process for determining which studies were eligible for inclusion.	
Data collection processes	10	Describe how IPD were requested, collected and managed, including any processes for querying and confirming data with investigators. If IPD were not sought from any eligible study, the reason for this should be stated (for each such study).	
		If applicable, describe how any studies for which IPD were not available were dealt with. This should include whether, how and what aggregate data were sought or extracted from study reports and publications (such as extracting data independently in duplicate) and any processes for obtaining and confirming these data with investigators.	
Data items	11	Describe how the information and variables to be collected were chosen. List and define all study level and participant level data that were sought, including baseline and follow-up information. If applicable, describe methods of standardising or translating variables within the IPD datasets to ensure common scales or measurements across studies.	
IPD integrity	A1	Describe what aspects of IPD were subject to data checking (such as sequence generation, data consistency and completeness, baseline imbalance) and how this was done.	
Risk of bias assessment in individual studies.	12	Describe methods used to assess risk of bias in the individual studies and whether this was applied separately for each outcome. If applicable, describe how findings of IPD checking were used to inform the assessment. Report if and how risk of bias assessment was used in any data synthesis.	
Specification of outcomes and effect measures	13	State all treatment comparisons of interests. State all outcomes addressed and define them in detail. State whether they were pre-specified for the review and, if applicable, whether they were primary/main or secondary/additional outcomes. Give the principal measures of effect (such as risk ratio, hazard ratio, difference in means) used for each outcome.	
Synthesis methods	14	Describe the meta-analysis methods used to synthesise IPD. Specify any statistical methods and models used. Issues should include (but are not restricted to): • Use of a one-stage or two-stage approach. • How effect estimates were generated separately within each study and combined across studies (where applicable). • Specification of one-stage models (where applicable) including how clustering of patients within studies was accounted for.	

		• Use of fixed or random effects models and any other model assumptions, such as proportional hazards. • How (summary) survival curves were generated (where applicable). • Methods for quantifying statistical heterogeneity (such as I^2 and τ^2). • How studies providing IPD and not providing IPD were analysed together (where applicable). • How missing data within the IPD were dealt with (where applicable).	
Exploration of variation in effects	A2	If applicable, describe any methods used to explore variation in effects by study or participant level characteristics (such as estimation of interactions between effect and covariates). State all participant-level characteristics that were analysed as potential effect modifiers, and whether these were pre-specified.	
Risk of bias across studies	15	Specify any assessment of risk of bias relating to the accumulated body of evidence, including any pertaining to not obtaining IPD for particular studies, outcomes or other variables.	
Additional analyses	16	Describe methods of any additional analyses, including sensitivity analyses. State which of these were pre-specified.	
Results			
Study selection and IPD obtained	17	Give numbers of studies screened, assessed for eligibility, and included in the systematic review with reasons for exclusions at each stage. Indicate the number of studies and participants for which IPD were sought and for which IPD were obtained. For those studies where IPD were not available, give the numbers of studies and participants for which aggregate data were available. Report reasons for non-availability of IPD. Include a flow diagram.	
Study characteristics	18	For each study, present information on key study and participant characteristics (such as description of interventions, numbers of participants, demographic data, unavailability of outcomes, funding source, and if applicable duration of follow-up). Provide (main) citations for each study. Where applicable, also report similar study characteristics for any studies not providing IPD.	
IPD integrity	A3	Report any important issues identified in checking IPD or state that there were none.	
Risk of bias within studies	19	Present data on risk of bias assessments. If applicable, describe whether data checking led to the up-weighting or down-weighting of these assessments. Consider how any potential bias impacts on the robustness of meta-analysis conclusions.	
Results of individual studies	20	For each comparison and for each main outcome (benefit or harm), for each individual study report the number of eligible participants for which data were obtained and show simple summary data for each intervention group (including, where applicable, the number of events), effect estimates and confidence intervals. These may be tabulated or included on a forest plot.	
Results of syntheses	21	Present summary effects for each meta-analysis undertaken, including confidence intervals and measures of statistical heterogeneity. State whether the analysis was pre-specified, and report the numbers of studies and participants and, where applicable, the number of events on which it is based.	

		When exploring variation in effects due to patient or study characteristics, present summary interaction estimates for each characteristic examined, including confidence intervals and measures of statistical heterogeneity. State whether the analysis was pre-specified. State whether any interaction is consistent across trials.	
		Provide a description of the direction and size of effect in terms meaningful to those who would put findings into practice.	
Risk of bias across studies	22	Present results of any assessment of risk of bias relating to the accumulated body of evidence, including any pertaining to the availability and representativeness of available studies, outcomes or other variables.	
Additional analyses	23	Give results of any additional analyses (e.g. sensitivity analyses). If applicable, this should also include any analyses that incorporate aggregate data for studies that do not have IPD. If applicable, summarise the main meta-analysis results following the inclusion or exclusion of studies for which IPD were not available.	
Discussion			
Summary of evidence	24	Summarise the main findings, including the strength of evidence for each main outcome.	
Strengths and limitations	25	Discuss any important strengths and limitations of the evidence including the benefits of access to IPD and any limitations arising from IPD that were not available.	
Conclusions	26	Provide a general interpretation of the findings in the context of other evidence.	
Implications	A4	Consider relevance to key groups (such as policy makers, service providers and service users). Consider implications for future research.	
Funding			
Funding	27	Describe sources of funding and other support (such as supply of IPD), and the role in the systematic review of those providing such support.	

A1 – A3 denote new items that are additional to standard PRISMA items. A4 has been created as a result of re-arranging content of the standard PRISMA statement to suit the way that systematic review IPD meta-analyses are reported.

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14. Annex 3: Search strategy in biomedical databases

14.1 MASCOT-1 PubMed search strategy

(((((non-pregnancy[All Fields] AND related[All Fields] AND ("infection"[MeSH Terms] OR "infection"[All Fields] OR "communicable diseases"[MeSH Terms] OR ("communicable"[All Fields] AND "diseases"[All Fields]) OR "communicable diseases"[All Fields])) OR non-pregnancy related[Title]) OR ((maternal[Title] OR pregnant[Title] OR pregnancy[Title] OR obstetric[Title] OR puerperal[Title] OR mother[Title] OR childbirth[Title] OR labour[Title] OR labor[Title] OR natal[Title] OR post-natal[Title] OR pre-natal[Title] OR prenatal[Title] OR antenatal[Title] OR ante-natal[Title] OR perinatal[Title] OR peri-natal[Title] OR puerperal[Title] OR puerperium[Title]) AND ((((((sepsis[Title] OR septic$[Title]) OR infection$[Title]) OR HIV[Title]) OR tuberculosis[Title]) OR pneumonia[Title]) OR meningitis[Title]))) OR (chorioamnionitis[Title/Abstract] OR "chorioamnionitis"[MeSH Terms])) OR ((("sepsis"[MeSH Terms] OR "sepsis"[All Fields]) OR septic$[All Fields] OR infection$[Title]) AND ((amniotic[Title/Abstract] OR intra-amniotic[Title/Abstract]) OR intraamniotic[Title/Abstract])))) OR ((((anemic[Title] OR anaemia[Title]) OR anaemic[Title]) OR anemia[Title]) AND (puerperal[Title] OR (((((maternal[Title] OR pregnant[Title]) OR pregnancy[Title]) OR obstetric[Title]) OR mother[Title]) OR childbirth[Title])))) OR ((((((("Midwifery"[Mesh] OR dula[Title/Abstract]) OR ((("parturition"[MeSH Terms] OR "parturition"[All Fields] OR "birth"[All Fields]) AND attendant[All Fields]) OR (("parturition"[MeSH Terms] OR "parturition"[All Fields] OR "birth"[All Fields]) AND attendants[All Fields]))) OR ("residence characteristics"[MeSH Terms] OR ("residence"[All Fields] AND "characteristics"[All Fields]) OR "residence characteristics"[All Fields] OR ("place"[All Fields] AND "birth"[All Fields]) OR "place of birth"[All Fields])) OR (("Birthing Centers"[MAJR] OR "Delivery Rooms"[MAJR]) OR "Delivery, Obstetric/nursing"[MAJR])) OR ((maternal[Title] OR pregnant[Title] OR pregnancy[Title] OR obstetric[Title] OR puerperal[Title] OR mother[Title] OR childbirth[Title] OR labour[Title] OR labor[Title] OR natal[Title] OR post-natal[Title] OR pre-natal[Title] OR prenatal[Title] OR antenatal[Title] OR ante-natal[Title] OR perinatal[Title] OR peri-natal[Title] OR puerperal[Title] OR puerperium[Title]) AND (("Ambulances"[Mesh] OR "Health Services Accessibility"[Mesh]) OR "Transportation of Patients"[Mesh]))) OR (("Travel"[MeSH Terms] OR "Delivery of Health Care/organization and administration"[MAJR]) AND (maternal[Title] OR pregnant[Title] OR pregnancy[Title] OR obstetric[Title] OR puerperal[Title] OR mother[Title] OR childbirth[Title] OR labour[Title] OR labor[Title] OR natal[Title] OR post-natal[Title] OR pre-natal[Title] OR prenatal[Title] OR antenatal[Title] OR ante-natal[Title] OR perinatal[Title] OR peri-natal[Title] OR puerperal[Title] OR puerperium[Title])))) OR (ectopic pregnancy[Title/Abstract] OR "pregnancy, ectopic"[MeSH Terms])) OR ((((((("Postpartum Hemorrhage"[Mesh] OR ((((((((((((((((maternal[Title] OR pregnant[Title]) OR pregnancy[Title]) OR obstetric[Title]) OR puerperal[Title]) OR mother[Title]) OR childbirth[Title]) OR labour[Title]) OR labor[Title]) OR natal[Title]) OR post-natal[Title]) OR pre-natal[Title]) OR prenatal[Title]) OR antenatal[Title]) OR ante-natal[Title]) OR perinatal[Title]) OR peri-natal[Title]) AND (Hemorrhage[Title] OR Haemorrhage[Title]))) OR (((obstetric[All Fields] AND ("haemorrhage"[All Fields] OR "hemorrhage"[MeSH Terms] OR "hemorrhage"[All Fields])) OR obstetric hemorrhage[Title/Abstract]) OR ("postpartum hemorrhage"[MeSH Terms] OR ("postpartum"[All Fields] AND "hemorrhage"[All Fields]) OR "postpartum hemorrhage"[All Fields] OR ("post"[All Fields] AND "partum"[All Fields] AND "hemorrhage"[All Fields]) OR "post

partum hemorrhage"[All Fields]) OR ("postpartum hemorrhage"[MeSH Terms] OR ("postpartum"[All Fields] AND "hemorrhage"[All Fields]) OR "postpartum hemorrhage"[All Fields] OR ("post"[All Fields] AND "partum"[All Fields] AND "haemorrhage"[All Fields]) OR "post partum haemorrhage"[All Fields]) OR ("postpartum hemorrhage"[MeSH Terms] OR ("postpartum"[All Fields] AND "hemorrhage"[All Fields]) OR "postpartum hemorrhage"[All Fields] OR ("post"[All Fields] AND "partum"[All Fields] AND "hemorrhage"[All Fields]) OR "post partum hemorrhage"[All Fields]) OR ("postpartum hemorrhage"[MeSH Terms] OR ("postpartum"[All Fields] AND "hemorrhage"[All Fields]) OR "postpartum hemorrhage"[All Fields] OR ("post"[All Fields] AND "partum"[All Fields] AND "haemorrhage"[All Fields]) OR "post partum haemorrhage"[All Fields])))) OR obstetric hemorrhage[Title/Abstract]) OR "Hypertension, Pregnancy-Induced"[Mesh]) OR ((((obstructed labor[Title/Abstract] OR obstructed labour[Title/Abstract]) OR (obstetric fistula[Title/Abstract] OR obstetric fistulae[Title/Abstract])) OR ("vaginal fistula"[MeSH Terms] OR "vesicovaginal fistula"[MeSH Terms])) OR ("Obstetric Labor Complications"[Mesh] OR "Obstetric Labor, Premature"[Mesh]))) OR ((((((((((((((((((maternal[Title] OR pregnant[Title]) OR pregnancy[Title]) OR obstetric[Title]) OR puerperal[Title]) OR mother[Title]) OR childbirth[Title]) OR labour[Title]) OR labor[Title]) OR natal[Title]) OR post-natal[Title]) OR pre-natal[Title]) OR prenatal[Title]) OR antenatal[Title]) OR ante-natal[Title]) OR perinatal[Title]) OR peri-natal[Title]) AND (hypertension[Title] OR blood pressure[Title])) AND (((((eclampsia[Title/Abstract] OR preeclampsia[Title/Abstract]) OR HELLP[Title/Abstract]) OR "eclampsia"[MeSH Terms]) OR "pre-eclampsia"[MeSH Terms]) OR pre-eclampsia[Title/Abstract])))) OR ("Pregnancy Complications, Hematologic"[Mesh] OR "Pregnancy in Adolescence"[Mesh] OR "Pregnancy Complications, Infectious"[Mesh] OR "Pregnancy Complications, Cardiovascular"[Mesh] OR "Pregnancy Complications"[Mesh] OR "Pregnancy, Prolonged"[Mesh])) AND ((((("africa"[MeSH Terms] OR "africa"[All Fields]) OR (((((((("afghanistan"[MeSH Terms] OR "afghanistan"[All Fields]) OR ("bangladesh"[MeSH Terms] OR "bangladesh"[All Fields])) OR ("benin"[MeSH Terms] OR "benin"[All Fields])) OR ("burkina faso"[MeSH Terms] OR ("burkina"[All Fields] AND "faso"[All Fields]) OR "burkina faso"[All Fields])) OR ((((((((((((((((((((((((("burundi"[MeSH Terms] OR "burundi"[All Fields]) OR ("cambodia"[MeSH Terms] OR "cambodia"[All Fields])) OR ("central african republic"[MeSH Terms] OR ("central"[All Fields] AND "african"[All Fields] AND "republic"[All Fields]) OR "central african republic"[All Fields])) OR ("chad"[MeSH Terms] OR "chad"[All Fields])) OR ("comoros"[MeSH Terms] OR "comoros"[All Fields])) OR (("congo"[MeSH Terms] OR "congo"[All Fields]) AND Dem.[All Fields] AND Rep[All Fields])) OR ("congo"[MeSH Terms] OR "congo"[All Fields])) OR DRC[Affiliation]) OR ("eritrea"[MeSH Terms] OR "eritrea"[All Fields])) OR ("ethiopia"[MeSH Terms] OR "ethiopia"[All Fields])) OR ("gambia"[MeSH Terms] OR "gambia"[All Fields])) OR ("guinea"[MeSH Terms] OR "guinea"[All Fields])) OR (("guinea"[MeSH Terms] OR "guinea"[All Fields]) AND Bisau[All Fields])) OR ("haiti"[MeSH Terms] OR "haiti"[All Fields])) OR ("kenya"[MeSH Terms] OR "kenya"[All Fields])) OR ("korea"[MeSH Terms] OR "korea"[All Fields])) OR Kyrgyz[All Fields]) OR ("liberia"[MeSH Terms] OR "liberia"[All Fields])) OR ("madagascar"[MeSH Terms] OR "madagascar"[All Fields])) OR ("malawi"[MeSH Terms] OR "malawi"[All Fields])) OR ("mali"[MeSH Terms] OR "mali"[All Fields])) OR ("mozambique"[MeSH Terms] OR "mozambique"[All Fields])) OR ("myanmar"[MeSH Terms] OR "myanmar"[All Fields])) OR ("nepal"[MeSH Terms] OR "nepal"[All Fields])) OR ("niger"[MeSH Terms] OR "niger"[All Fields])) OR ("rwanda"[MeSH Terms] OR "rwanda"[All Fields])) OR ("sierra leone"[MeSH Terms] OR ("sierra"[All Fields] AND "leone"[All Fields]) OR "sierra leone"[All Fields])) OR ("somalia"[MeSH Terms] OR "somalia"[All Fields]))) OR (((((("tajikistan"[MeSH Terms] OR "tajikistan"[All Fields]) OR ("tanzania"[MeSH Terms] OR "tanzania"[All Fields])) OR ("togo"[MeSH Terms] OR "togo"[All Fields])) OR ("uganda"[MeSH Terms] OR "uganda"[All Fields])) OR ("zimbabwe"[MeSH Terms] OR "zimbabwe"[All Fields]))) OR ("africa, northern"[MeSH Terms] OR ("africa"[All Fields] AND

"northern"[All Fields]) OR "northern africa"[All Fields] OR "sahara"[All Fields])) OR sub-saharan[All Fields])) OR ((("angola"[MeSH Terms] OR "angola"[All Fields]) OR ("armenia"[MeSH Terms] OR "armenia"[All Fields])) OR ("belize"[MeSH Terms] OR "belize"[All Fields])) OR ("bhutan"[MeSH Terms] OR "bhutan"[All Fields])) OR ("bolivia"[MeSH Terms] OR "bolivia"[All Fields])) OR ("cameroon"[MeSH Terms] OR "cameroon"[All Fields])) OR ("cape verde"[MeSH Terms] OR ("cape"[All Fields] AND "verde"[All Fields]) OR "cape verde"[All Fields])) OR ("congo"[MeSH Terms] OR "congo"[All Fields])) OR ("cote d'ivoire"[MeSH Terms] OR ("cote"[All Fields] AND "d'ivoire"[All Fields]) OR "cote d'ivoire"[All Fields])) OR ("cote d'ivoire"[MeSH Terms] OR ("cote"[All Fields] AND "d'ivoire"[All Fields]) OR "cote d'ivoire"[All Fields] OR ("ivory"[All Fields] AND "coast"[All Fields]) OR "ivory coast"[All Fields])) OR ("djibouti"[MeSH Terms] OR "djibouti"[All Fields])) OR ("egypt"[MeSH Terms] OR "egypt"[All Fields])) OR ("el salvador"[MeSH Terms] OR ("el"[All Fields] AND "salvador"[All Fields]) OR "el salvador"[All Fields])) OR ("fiji"[MeSH Terms] OR "fiji"[All Fields])) OR ("georgia"[MeSH Terms] OR "georgia"[All Fields] OR "georgia (republic)"[MeSH Terms] OR ("georgia"[All Fields] AND "(republic)"[All Fields]) OR "georgia (republic)"[All Fields])) OR ("ghana"[MeSH Terms] OR "ghana"[All Fields])) OR ("guatemala"[MeSH Terms] OR "guatemala"[All Fields])) OR ("guyana"[MeSH Terms] OR "guyana"[All Fields])) OR ("honduras"[MeSH Terms] OR "honduras"[All Fields])) OR ("indonesia"[MeSH Terms] OR "indonesia"[All Fields])) OR ("india"[MeSH Terms] OR "india"[All Fields])) OR ("iraq"[MeSH Terms] OR "iraq"[All Fields])) OR ("micronesia"[MeSH Terms] OR "micronesia"[All Fields] OR "kiribati"[All Fields])) OR ("yugoslavia"[MeSH Terms] OR "yugoslavia"[All Fields] OR "kosovo"[All Fields])) OR Lao[All Fields]) OR ("lesotho"[MeSH Terms] OR "lesotho"[All Fields])) OR ("micronesia"[MeSH Terms] OR "micronesia"[All Fields] OR ("marshall"[All Fields] AND "islands"[All Fields]) OR "marshall islands"[All Fields])) OR ("mauritania"[MeSH Terms] OR "mauritania"[All Fields])) OR ("micronesia"[MeSH Terms] OR "micronesia"[All Fields])) OR ("moldova"[MeSH Terms] OR "moldova"[All Fields])) OR ("mongolia"[MeSH Terms] OR "mongolia"[All Fields])) OR ("morocco"[MeSH Terms] OR "morocco"[All Fields])) OR ("nicaragua"[MeSH Terms] OR "nicaragua"[All Fields])) OR ("nigeria"[MeSH Terms] OR "nigeria"[All Fields])) OR ("pakistan"[MeSH Terms] OR "pakistan"[All Fields])) OR ("papua new guinea"[MeSH Terms] OR ("papua"[All Fields] AND "new"[All Fields] AND "guinea"[All Fields]) OR "papua new guinea"[All Fields])) OR ("paraguay"[MeSH Terms] OR "paraguay"[All Fields])) OR ("philippines"[MeSH Terms] OR "philippines"[All Fields])) OR ("samoa"[MeSH Terms] OR "samoa"[All Fields])) OR ("atlantic islands"[MeSH Terms] OR ("atlantic"[All Fields] AND "islands"[All Fields]) OR "atlantic islands"[All Fields] OR ("sao"[All Fields] AND "tome"[All Fields] AND "principe"[All Fields]) OR "sao tome and principe"[All Fields])) OR (Sao[All Fields] AND Tome[All Fields])) OR ("senegal"[MeSH Terms] OR "senegal"[All Fields])) OR ("melanesia"[MeSH Terms] OR "melanesia"[All Fields] OR ("solomon"[All Fields] AND "islands"[All Fields]) OR "solomon islands"[All Fields])) OR ("sri lanka"[MeSH Terms] OR ("sri"[All Fields] AND "lanka"[All Fields]) OR "sri lanka"[All Fields])) OR ("sudan"[MeSH Terms] OR "sudan"[All Fields])) OR ("swaziland"[MeSH Terms] OR "swaziland"[All Fields])) OR ("syria"[MeSH Terms] OR "syria"[All Fields] OR ("syrian"[All Fields] AND "arab"[All Fields] AND "republic"[All Fields]) OR "syrian arab republic"[All Fields])) OR ("syria"[MeSH Terms] OR "syria"[All Fields])) OR ("east timor"[MeSH Terms] OR ("east"[All Fields] AND "timor"[All Fields]) OR "east timor"[All Fields] OR ("timor"[All Fields] AND "leste"[All Fields]) OR "timor leste"[All Fields])) OR ("indonesia"[MeSH Terms] OR "indonesia"[All Fields] OR "timor"[All Fields])) OR ("tonga"[MeSH Terms] OR "tonga"[All Fields])) OR ("turkmenistan"[MeSH Terms] OR "turkmenistan"[All Fields])) OR ("micronesia"[MeSH Terms] OR "micronesia"[All Fields] OR "tuvalu"[All Fields])) OR ("ukraine"[MeSH Terms] OR "ukraine"[All Fields])) OR ("uzbekistan"[MeSH Terms] OR "uzbekistan"[All Fields])) OR ("vanuatu"[MeSH Terms] OR "vanuatu"[All Fields])) OR ("vietnam"[MeSH Terms] OR "vietnam"[All Fields])) OR (("middle east"[MeSH Terms] OR

("middle"[All Fields] AND "east"[All Fields]) OR "middle east"[All Fields] OR ("west"[All Fields] AND "bank"[All Fields]) OR "west bank"[All Fields]) AND Gaza[All Fields])) OR Gaza[All Fields]) OR ("yemen"[MeSH Terms] OR "yemen"[All Fields])) OR ("zambia"[MeSH Terms] OR "zambia"[All Fields]))) OR ((("albania"[MeSH Terms] OR "albania"[All Fields]) OR ("algeria"[MeSH Terms] OR "algeria"[All Fields])) OR ("american samoa"[MeSH Terms] OR ("american"[All Fields] AND "samoa"[All Fields]) OR "american samoa"[All Fields])) OR ("samoa"[MeSH Terms] OR "samoa"[All Fields])) OR ("antigua and barbuda"[MeSH Terms] OR ("antigua"[All Fields] AND "barbuda"[All Fields]) OR "antigua and barbuda"[All Fields])) OR ("antigua and barbuda"[MeSH Terms] OR ("antigua"[All Fields] AND "barbuda"[All Fields]) OR "antigua and barbuda"[All Fields] OR "antigua"[All Fields])) OR ("antigua and barbuda"[MeSH Terms] OR ("antigua"[All Fields] AND "barbuda"[All Fields]) OR "antigua and barbuda"[All Fields] OR "barbuda"[All Fields])) OR ("argentina"[MeSH Terms] OR "argentina"[All Fields])) OR ("azerbaijan"[MeSH Terms] OR "azerbaijan"[All Fields])) OR ("republic of belarus"[MeSH Terms] OR ("republic"[All Fields] AND "belarus"[All Fields]) OR "republic of belarus"[All Fields] OR "belarus"[All Fields])) OR ("bosnia-herzegovina"[MeSH Terms] OR "bosnia-herzegovina"[All Fields] OR ("bosnia"[All Fields] AND "herzegovina"[All Fields]) OR "bosnia and herzegovina"[All Fields])) OR ("bosnia-herzegovina"[MeSH Terms] OR "bosnia-herzegovina"[All Fields] OR "bosnia"[All Fields])) OR ("bosnia-herzegovina"[MeSH Terms] OR "bosnia-herzegovina"[All Fields] OR "herzegovina"[All Fields])) OR ("botswana"[MeSH Terms] OR "botswana"[All Fields])) OR ("brazil"[MeSH Terms] OR "brazil"[All Fields])) OR ("bulgaria"[MeSH Terms] OR "bulgaria"[All Fields])) OR ("chile"[MeSH Terms] OR "chile"[All Fields])) OR ("china"[MeSH Terms] OR "china"[All Fields])) OR ("colombia"[MeSH Terms] OR "colombia"[All Fields])) OR ("costa rica"[MeSH Terms] OR ("costa"[All Fields] AND "rica"[All Fields]) OR "costa rica"[All Fields])) OR ("cuba"[MeSH Terms] OR "cuba"[All Fields])) OR ("dominica"[MeSH Terms] OR "dominica"[All Fields])) OR ("dominican republic"[MeSH Terms] OR ("dominican"[All Fields] AND "republic"[All Fields]) OR "dominican republic"[All Fields])) OR ("ecuador"[MeSH Terms] OR "ecuador"[All Fields])) OR ("gabon"[MeSH Terms] OR "gabon"[All Fields])) OR ("grenada"[MeSH Terms] OR "grenada"[All Fields])) OR ("iran"[MeSH Terms] OR "iran"[All Fields])) OR ("jamaica"[MeSH Terms] OR "jamaica"[All Fields])) OR ("jordan"[MeSH Terms] OR "jordan"[All Fields])) OR ("kazakhstan"[MeSH Terms] OR "kazakhstan"[All Fields])) OR ("latvia"[MeSH Terms] OR "latvia"[All Fields])) OR ("lebanon"[MeSH Terms] OR "lebanon"[All Fields])) OR ("libya"[MeSH Terms] OR "libya"[All Fields])) OR ("lithuania"[MeSH Terms] OR "lithuania"[All Fields])) OR ("macedonia (republic)"[MeSH Terms] OR ("macedonia"[All Fields] AND "(republic)"[All Fields]) OR "macedonia (republic)"[All Fields] OR "macedonia"[All Fields])) OR ("malaysia"[MeSH Terms] OR "malaysia"[All Fields])) OR ("indian ocean islands"[MeSH Terms] OR ("indian"[All Fields] AND "ocean"[All Fields] AND "islands"[All Fields]) OR "indian ocean islands"[All Fields] OR "maldives"[All Fields])) OR ("mauritius"[MeSH Terms] OR "mauritius"[All Fields])) OR ("comoros"[MeSH Terms] OR "comoros"[All Fields] OR "mayotte"[All Fields])) OR ("mexico"[MeSH Terms] OR "mexico"[All Fields])) OR ("montenegro"[MeSH Terms] OR "montenegro"[All Fields])) OR ("namibia"[MeSH Terms] OR "namibia"[All Fields])) OR ("palau"[MeSH Terms] OR "palau"[All Fields])) OR ("panama"[MeSH Terms] OR "panama"[All Fields])) OR ("peru"[MeSH Terms] OR "peru"[All Fields])) OR ("romania"[MeSH Terms] OR "romania"[All Fields])) OR ("russia"[MeSH Terms] OR "russia"[All Fields] OR ("russian"[All Fields] AND "federation"[All Fields]) OR "russian federation"[All Fields])) OR ("russia"[MeSH Terms] OR "russia"[All Fields])) OR ("ussr"[MeSH Terms] OR "ussr"[All Fields])) OR ("serbia"[MeSH Terms] OR "serbia"[All Fields])) OR ("seychelles"[MeSH Terms] OR "seychelles"[All Fields])) OR ("south africa"[MeSH Terms] OR ("south"[All Fields] AND "africa"[All Fields]) OR "south africa"[All Fields])) OR ("saint kitts and nevis"[MeSH Terms] OR ("saint"[All Fields] AND "kitts"[All Fields] AND "nevis"[All Fields]) OR "saint kitts and nevis"[All Fields] OR ("st"[All Fields] AND "kitts"[All Fields] AND "nevis"[All Fields])

OR "st kitts and nevis"[All Fields])) OR ("saint kitts and nevis"[MeSH Terms] OR ("saint"[All Fields] AND "kitts"[All Fields] AND "nevis"[All Fields]) OR "saint kitts and nevis"[All Fields])) OR (("saints"[MeSH Terms] OR "saints"[All Fields] OR "saint"[All Fields]) AND Kitts[All Fields])) OR ("saint lucia"[MeSH Terms] OR ("saint"[All Fields] AND "lucia"[All Fields]) OR "saint lucia"[All Fields] OR ("st"[All Fields] AND "lucia"[All Fields]) OR "st lucia"[All Fields])) OR ("saint lucia"[MeSH Terms] OR ("saint"[All Fields] AND "lucia"[All Fields]) OR "saint lucia"[All Fields])) OR ("saint vincent and the grenadines"[MeSH Terms] OR ("saint"[All Fields] AND "vincent"[All Fields] AND "grenadines"[All Fields]) OR "saint vincent and the grenadines"[All Fields] OR ("st"[All Fields] AND "vincent"[All Fields] AND "grenadines"[All Fields]) OR "st vincent and the grenadines"[All Fields])) OR (St. Vincent[Author] OR St. Vincent[Investigator])) OR ("saint vincent and the grenadines"[MeSH Terms] OR ("saint"[All Fields] AND "vincent"[All Fields] AND "grenadines"[All Fields]) OR "saint vincent and the grenadines"[All Fields])) OR Saint Vincent[Author]) OR ("saint vincent and the grenadines"[MeSH Terms] OR ("saint"[All Fields] AND "vincent"[All Fields] AND "grenadines"[All Fields]) OR "saint vincent and the grenadines"[All Fields] OR "grenadines"[All Fields])) OR ("suriname"[MeSH Terms] OR "suriname"[All Fields])) OR ("thailand"[MeSH Terms] OR "thailand"[All Fields])) OR ("tunisia"[MeSH Terms] OR "tunisia"[All Fields])) OR ("turkey"[MeSH Terms] OR "turkey"[All Fields])) OR ("uruguay"[MeSH Terms] OR "uruguay"[All Fields])) OR ("venezuela"[MeSH Terms] OR "venezuela"[All Fields])))

14.2 MASCOT-2 PubMed search strategy

Full search 7 October 2020

2012/09/01:2020/10/07[Date - Create] AND (("cohort studies"[MeSH Terms:noexp] OR "longitudinal studies"[MeSH Terms:noexp] OR "follow up studies"[MeSH Terms:noexp] OR "prospective studies"[MeSH Terms:noexp] OR "retrospective studies"[MeSH Terms:noexp] OR "cohort"[Title/Abstract] OR "longitudinal"[Title/Abstract] OR "prospective"[Title/Abstract] OR "retrospective"[Title/Abstract] OR ((("Clinical Trial"[Publication Type] OR "Clinical Trials as Topic"[MeSH Terms] OR "clinical trials"[All Fields]) AND "Clinical Trial"[Publication Type:noexp]) OR "clinical trial, phase i"[Publication Type] OR "clinical trial, phase ii"[Publication Type] OR "clinical trial, phase iii"[Publication Type] OR "clinical trial, phase iv"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "multicenter study"[Publication Type] OR "randomized controlled trial"[Publication Type] OR "Clinical Trials as Topic"[MeSH Terms:noexp] OR "clinical trials, phase i as topic"[MeSH Terms:noexp] OR "clinical trials, phase ii as topic"[MeSH Terms:noexp] OR "clinical trials, phase iii as topic"[MeSH Terms:noexp] OR "clinical trials, phase iv as topic"[MeSH Terms:noexp] OR "controlled clinical trials as topic"[MeSH Terms:noexp] OR "randomized controlled trials as topic"[MeSH Terms:noexp] OR "early termination of clinical trials"[MeSH Terms:noexp] OR "multicenter studies as topic"[MeSH Terms:noexp] OR "Double-Blind Method"[MeSH Terms] OR (("randomised"[Title/Abstract] OR "randomized"[Title/Abstract]) AND ("trial"[Title/Abstract] OR "trials"[Title/Abstract])) OR (("single"[Title/Abstract] OR "double"[Title/Abstract] OR "doubled"[Title/Abstract] OR "triple"[Title/Abstract] OR "tripled"[Title/Abstract] OR "treble"[Title/Abstract] OR "treble"[Title/Abstract]) AND ("blind*"[Title/Abstract] OR "mask*"[Title/Abstract])) OR ("4 arm"[Title/Abstract] OR "four arm"[Title/Abstract]) OR "incidence"[MeSH Terms:noexp] OR "incidence"[Title/Abstract])) AND ("angola"[MeSH Terms] OR "benin"[MeSH Terms] OR "botswana"[MeSH Terms] OR "burkina faso"[MeSH Terms] OR "burundi"[MeSH Terms] OR "cabo verde"[MeSH Terms] OR "cameroon"[MeSH Terms] OR "central african republic"[MeSH Terms] OR "chad"[MeSH Terms] OR "comoros"[MeSH Terms] OR "democratic republic of the congo"[MeSH Terms] OR "congo"[MeSH Terms] OR "cote d ivoire"[MeSH Terms] OR "djibouti"[MeSH Terms] OR "equatorial guinea"[MeSH Terms] OR "eritrea"[MeSH Terms] OR "eswatini"[MeSH Terms] OR "ethiopia"[MeSH Terms] OR "gabon"[MeSH Terms] OR "gambia"[MeSH Terms] OR "ghana"[MeSH Terms] OR "guinea"[MeSH Terms] OR "guinea bissau"[MeSH Terms] OR "kenya"[MeSH Terms] OR

"lesotho"[MeSH Terms] OR "madagascar"[MeSH Terms] OR "malawi"[MeSH Terms] OR "mali"[MeSH Terms] OR "mauritania"[MeSH Terms] OR "mozambique"[MeSH Terms] OR "namibia"[MeSH Terms] OR "niger"[MeSH Terms] OR "nigeria"[MeSH Terms] OR "rwanda"[MeSH Terms] OR "senegal"[MeSH Terms] OR "sierra leone"[MeSH Terms] OR "somalia"[MeSH Terms] OR "south africa"[MeSH Terms] OR "south sudan"[MeSH Terms] OR "sudan"[MeSH Terms] OR "tanzania"[MeSH Terms] OR "togo"[MeSH Terms] OR "uganda"[MeSH Terms] OR "zambia"[MeSH Terms] OR "zimbabwe"[MeSH Terms] OR "africa south of the sahara"[MeSH Terms] OR "africa, central"[MeSH Terms] OR "africa, southern"[MeSH Terms] OR "africa, eastern"[MeSH Terms] OR "africa, western"[MeSH Terms] OR ("angola"[Text Word] OR "benin"[Text Word] OR "botswana"[Text Word] OR "bechuanaland"[Text Word] OR "burkina faso"[Text Word] OR "burkina fasso"[Text Word] OR "upper volta"[Text Word] OR "burundi"[Text Word] OR "urundi"[Text Word] OR "cabo verde"[Text Word] OR "cape verde"[Text Word] OR 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OR "nigeria"[Text Word] OR "rwanda"[Text Word] OR "ruanda"[Text Word] OR "senegal"[Text Word] OR "sierra leone"[Text Word] OR "somalia"[Text Word] OR "south africa"[Text Word] OR "south sudan"[Text Word] OR "sudan"[Text Word] OR "tanzania"[Text Word] OR "tanganyika"[Text Word] OR "togo"[Text Word] OR "togolese republic"[Text Word] OR "tonga"[Text Word] OR "uganda"[Text Word] OR "zambia"[Text Word] OR "zimbabwe"[Text Word] OR "northern rhodesia"[Text Word] OR "africa south of the sahara"[Text Word] OR "sub saharan africa"[Text Word] OR "subsaharan africa"[Text Word] OR "central africa"[Text Word] OR "sahara"[Text Word] OR "southern africa"[Text Word] OR "east africa"[Text Word] OR "eastern africa"[Text Word] OR "west africa"[Text Word] OR "western africa"[Text Word])) AND (("non-pregnancy"[All Fields] AND ("family"[MeSH Terms] OR "family"[All Fields] OR "relation"[All Fields] OR "relatability"[All Fields] OR "relatable"[All Fields] OR "related"[All Fields] OR "relates"[All Fields] OR "relating"[All Fields] OR "relational"[All Fields] OR "relations"[All Fields]) AND ("infections"[MeSH Terms] OR "infection"[All Fields] OR "communicable diseases"[MeSH Terms] OR ("communicable"[All Fields] AND "diseases"[All Fields]) OR "communicable diseases"[All Fields])) OR "non pregnancy related"[Title] OR (("maternal"[Title] OR "pregnant"[Title] OR "pregnancy"[Title] OR "obstetric"[Title] OR "puerperal"[Title] OR "mother"[Title] OR "childbirth"[Title] OR "labour"[Title] OR "labor"[Title] OR "natal"[Title] OR "post-natal"[Title] OR "pre-natal"[Title] OR "prenatal"[Title] OR "antenatal"[Title] OR "ante-natal"[Title] OR "perinatal"[Title] OR "peri-natal"[Title] OR "puerperal"[Title] OR "puerperium"[Title]) AND ("sepsis"[Title] OR "septic*"[Title] OR "infection*"[Title] OR "HIV"[Title] OR "tuberculosis"[Title] OR "pneumonia"[Title] OR "meningitis"[Title])) OR ("chorioamnionitis"[Title/Abstract] OR "chorioamnionitis"[MeSH Terms]) OR (("sepsis"[MeSH Terms] OR "sepsis"[All Fields] OR "septic*"[All Fields] OR "infection*"[Title]) AND ("amniotic"[Title/Abstract] OR "intra-amniotic"[Title/Abstract] OR "intraamniotic"[Title/Abstract])) OR (("anemic"[Title] OR "anaemia"[Title] OR "anaemic"[Title] OR "anemia"[Title]) AND ("puerperal"[Title] OR ("maternal"[Title] OR "pregnant"[Title] OR "pregnancy"[Title] OR "obstetric"[Title] OR "mother"[Title] OR "childbirth"[Title]))) OR ("Midwifery"[MeSH Terms] OR "dula"[Title/Abstract] OR ((("parturition"[MeSH Terms] OR "parturition"[All Fields] OR "birth"[All Fields]) AND ("attend"[All Fields] OR "attendance"[All Fields] OR "attendances"[All Fields] OR "attendant"[All Fields] OR "attendant s"[All Fields] OR "attendants"[All Fields] OR "attended"[All Fields] OR "attendence"[All Fields] OR "attendents"[All Fields] OR "attender"[All Fields] OR "attenders"[All Fields] OR "attending"[All Fields] OR "attendings"[All Fields] OR "attends"[All Fields])) OR (("parturition"[MeSH Terms] OR "parturition"[All Fields] OR "birth"[All Fields]) AND ("attend"[All Fields] OR "attendance"[All Fields] OR "attendances"[All Fields] OR "attendant"[All Fields] OR "attendant s"[All Fields] OR "attendants"[All Fields] OR "attended"[All Fields] OR "attendence"[All Fields] OR "attendents"[All Fields] OR "attender"[All Fields] OR "attenders"[All Fields]

OR "attending"[All Fields] OR "attendings"[All Fields] OR "attends"[All Fields])) OR ("residence characteristics"[MeSH Terms] OR ("residence"[All Fields] AND "characteristics"[All Fields]) OR "residence characteristics"[All Fields] OR ("place"[All Fields] AND "birth"[All Fields]) OR "place of birth"[All Fields]) OR ("Birthing Centers"[MeSH Major Topic] OR "Delivery Rooms"[MeSH Major Topic] OR "delivery, obstetric/nursing"[MeSH Major Topic]) OR (("maternal"[Title] OR "pregnant"[Title] OR "pregnancy"[Title] OR "obstetric"[Title] OR "puerperal"[Title] OR "mother"[Title] OR "childbirth"[Title] OR "labour"[Title] OR "labor"[Title] OR "natal"[Title] OR "post-natal"[Title] OR "pre-natal"[Title] OR "prenatal"[Title] OR "antenatal"[Title] OR "ante-natal"[Title] OR "perinatal"[Title] OR "peri-natal"[Title] OR "puerperal"[Title] OR "puerperium"[Title]) AND ("Ambulances"[MeSH Terms] OR "Health Services Accessibility"[MeSH Terms] OR "Transportation of Patients"[MeSH Terms])) OR (("Travel"[MeSH Terms] OR "delivery of health care/organization and administration"[MeSH Major Topic]) AND ("maternal"[Title] OR "pregnant"[Title] OR "pregnancy"[Title] OR "obstetric"[Title] OR "puerperal"[Title] OR "mother"[Title] OR "childbirth"[Title] OR "labour"[Title] OR "labor"[Title] OR "natal"[Title] OR "post-natal"[Title] OR "pre-natal"[Title] OR "prenatal"[Title] OR "antenatal"[Title] OR "ante-natal"[Title] OR "perinatal"[Title] OR "peri-natal"[Title] OR "puerperal"[Title] OR "puerperium"[Title]))) OR ("ectopic pregnancy"[Title/Abstract] OR "pregnancy, ectopic"[MeSH Terms]) OR ("Postpartum Hemorrhage"[MeSH Terms] OR (("maternal"[Title] OR "pregnant"[Title] OR "pregnancy"[Title] OR "obstetric"[Title] OR "puerperal"[Title] OR "mother"[Title] OR "childbirth"[Title] OR "labour"[Title] OR "labor"[Title] OR "natal"[Title] OR "post-natal"[Title] OR "pre-natal"[Title] OR "prenatal"[Title] OR "antenatal"[Title] OR "ante-natal"[Title] OR "perinatal"[Title] OR "peri-natal"[Title]) AND ("Hemorrhage"[Title] OR "Haemorrhage"[Title])) OR ((("obstetric"[All Fields] OR "obstetrically"[All Fields] OR "obstetrics"[MeSH Terms] OR "obstetrics"[All Fields] OR "obstetrical"[All Fields]) AND ("Haemorrhage"[All Fields] OR "Hemorrhage"[MeSH Terms] OR "Hemorrhage"[All Fields])) OR "obstetric hemorrhage"[Title/Abstract] OR ("Postpartum Hemorrhage"[MeSH Terms] OR ("postpartum"[All Fields] AND "Hemorrhage"[All Fields]) OR "Postpartum Hemorrhage"[All Fields] OR ("post"[All Fields] AND "partum"[All Fields] AND "Hemorrhage"[All Fields]) OR "post partum hemorrhage"[All Fields]) OR ("Postpartum Hemorrhage"[MeSH Terms] OR ("postpartum"[All Fields] AND "Hemorrhage"[All Fields]) OR "Postpartum Hemorrhage"[All Fields] OR ("post"[All Fields] AND "partum"[All Fields] AND "Haemorrhage"[All Fields]) OR "post partum haemorrhage"[All Fields]) OR ("Postpartum Hemorrhage"[MeSH Terms] OR ("postpartum"[All Fields] AND "Hemorrhage"[All Fields]) OR "Postpartum Hemorrhage"[All Fields] OR ("post"[All Fields] AND "partum"[All Fields] AND "Hemorrhage"[All Fields]) OR "post partum hemorrhage"[All Fields]) OR ("Postpartum Hemorrhage"[MeSH Terms] OR ("postpartum"[All Fields] AND "Hemorrhage"[All Fields]) OR "Postpartum Hemorrhage"[All Fields] OR ("post"[All Fields] AND "partum"[All Fields] AND "Haemorrhage"[All Fields]) OR "post partum haemorrhage"[All Fields])) OR "obstetric hemorrhage"[Title/Abstract] OR "hypertension, pregnancy induced"[MeSH Terms] OR ("obstructed labor"[Title/Abstract] OR "obstructed labour"[Title/Abstract] OR ("obstetric fistula"[Title/Abstract] OR "obstetric fistulae"[Title/Abstract]) OR ("vaginal fistula"[MeSH Terms] OR "vesicovaginal fistula"[MeSH Terms]) OR ("Obstetric Labor Complications"[MeSH Terms] OR "obstetric labor, premature"[MeSH Terms])) OR (("maternal"[Title] OR "pregnant"[Title] OR "pregnancy"[Title] OR "obstetric"[Title] OR "puerperal"[Title] OR "mother"[Title] OR "childbirth"[Title] OR "labour"[Title] OR "labor"[Title] OR "natal"[Title] OR "post-natal"[Title] OR "pre-natal"[Title] OR "prenatal"[Title] OR "antenatal"[Title] OR "ante-natal"[Title] OR "perinatal"[Title] OR "peri-natal"[Title]) AND ("hypertension"[Title] OR "blood pressure"[Title]) AND ("eclampsia"[Title/Abstract] OR "preeclampsia"[Title/Abstract] OR "HELLP"[Title/Abstract] OR "eclampsia"[MeSH Terms] OR "pre-eclampsia"[MeSH Terms] OR "pre-eclampsia"[Title/Abstract]))) OR ("pregnancy complications, hematologic"[MeSH Terms] OR "Pregnancy in Adolescence"[MeSH Terms] OR "pregnancy complications, infectious"[MeSH Terms] OR "pregnancy complications, cardiovascular"[MeSH Terms] OR "Pregnancy Complications"[MeSH Terms] OR "pregnancy, prolonged"[MeSH Terms])))

15. Annex 4: Participant level variables

		-	-
			•
	•	•	
	•	•	•
Variables	Variables essential for study inclusion	Important variables	Desirable variables
Maternal outcomes	To be eligible, data should be available on at least 2 of the important maternal or neonatal outcome variables	• Gestational age at delivery • Preterm premature rupture of the membranes (PPROM) • Prolonged rupture of membranes (PROM) • Antepartum and postpartum hemorrhage estimated blood loss • Hypertensive disorders in pregnancy - gestational hypertension - preeclampsia - preeclampsia with severe features - eclampsia - HELLP syndrome - blood pressure (systolic/diastolic) - proteinuria • Anaemia in pregnancy - hemoglobin, mean cellular volume, hematocrit	• Duration of labor • Caesarean section - emergency - elective • Abortion - spontaneous (miscarriage) - threatened spontaneous - induced • Oligohydramnios • Placental complications - placental abruption - placenta previa - fetal growth restriction • Sexual and gender-based violence - intimate partner violence • Maternal mortality (including cause) • Hyperemesis gravidarum

		• Adverse events - serious adverse events (SAEs) • Gestational Diabetes Mellitus (GDM) - glucose level in pregnancy (hemoglobin A1c) - Oral Glucose Tolerance Test (OGTT) • Health facility visits - emergency department visits - hospital admissions • Maternal mental health - emotional stress, maternal global severity index (GSI) - Life Event Scale for Pregnant Women - Patient Health Questionnaire-9 (PHQ-9) - Other validated scales/measures	• Maternal cardiovascular disease - ischemic heart disease - stroke - heart failure • Cardiac arrest • Renal function - Glomerular filtration rate (GFR), urea, creatinine • Liver function - ALAT, ASAT, total bilirubin/conjugated bilirubin, GGT • Maternal peripartum infections - pyelonephritis - puerperal sepsis - chorioamnionitis - Group B streptococcus - urinary tract infection/bacteriuria • Infectious disease - malaria - dengue - TB - Other • Maternal immunization • Maternal caregiving practices • Ectopic pregnancy • HIV status - newly diagnosed, chronic - CD4, viral load - treatment
Fetal, neonatal and child outcomes	To be eligible, data should be available on at least 2 of the important maternal or neonatal outcome variables	• Prematurity (see also gestational age at delivery) • Mortality (including cause) - stillbirth (fresh/macerated) - neonatal - perinatal - child (first two years)	• Birth - singleton/multiple • Meconium staining • Infant sex • Infant feeding practices - exclusive breastfeeding (if yes, duration)

		• Mother-to-child transmission of HIV (MTCT) • APGAR score • Infant growth - small for gestational age - infant height (<2 years) - failure to thrive - stunting • Admission to neonatal intensive care units or paediatric ward • Intrauterine growth restriction • Ultrasound findings	• Fetal distress, hypoxia • Infections - neonatal sepsis - TORCH (toxoplasmosis, other (syphilis, hepatitis B), rubella, cytomegalovirus, herpes simplex) - group B streptococcus - respiratory tract infection (lower/upper/pneumonia) - diarrhea • Early Child Development (ECD) • Bayley's score (<2 years) • Neonatal jaundice • serum and/or transcutaneous bilirubin levels Congenital anomaly
Other variables	To be eligible, data should be available on: • Date of delivery of the newborn OR date of maternal outcome • Location, at a minimum: city of delivery, or city of follow-up (data on location of household, birth facility, or study clinic are preferable)	• Maternal age • Gravidity, parity • Study intervention or exposure • Maternal anthropometry - Maternal weight, height, BMI, MUAC • Date of interviews or examination • Mode of delivery • Facility of delivery location, or catchment area of facility • Location of research site • Type of facility (community health center/hospital) -	• Time of delivery • Location o home address o rural/urban/peri-urban • Housing type o apartment, house, informal o no. of people in household o air-conditioning access • Socio-economic status or income o personal income o household income • Race, ethnicity • Substance use o Smoking, alcohol, or illicit substances • Employment status • Maternal co-morbidities o chronic medication • Education (highest level achieved) • Marital status

			• Birth attendant (skilled/unskilled) • Religion • Lost to follow-up • Temperature in healthcare facility, incubator, crib, room

Index	Index Name	Longer description
HI	Heat Index	Heat index (HI) is a heat stress indicator used by the US National Oceanic and Atmospheric Administration (NOAA) National Weather Service for issuing heat warnings. It is calculated using multiple linear regression based on daily maximum temperature and relative humidity (calculated from daily mean specific humidity and surface pressure). Unit: deg C. Valid range: -100-100
Hu	Humidex	Humidex (Hu) is a heat stress indicator used by Canadian meteorological services. It is calculated as linear combination of daily maximum temperature and vapour pressure (calculated from daily mean specific humidity and surface pressure). Unit: °C. Valid range: -100-100.
UTCI	Universal Thermal Climate Index	UTCI is a heat stress indicator that was developed from a multi-node model of human heat transfer. Here it is calculated as polynomial approximation to this model based on daily maximum temperature and vapour pressure (calculated from daily mean specific humidity and surface pressure). Unit: °C. Valid range: -100-100
WBT	The wet-bulb temperature index	The wet-bulb temperature index is a heat stress indicator that indicates the human cooling capacity through sweating. It is calculated from equivalent potential temperature based on daily maximum temperature and water vapour mixing ratio (calculated from daily mean specific humidity and surface pressure). Unit: °C. Valid range: -100-100
WBGT	The wet-bulb globe temperature index	The wet-bulb globe temperature index is a heat stress indicator calculated as weighted mean of wet-bulb temperature and daily maximum temperature. Unit: °C. Valid range: -100-100.
DI	Discomfort Index	Discomfort index as per Thom (1959) and Sohar (1963). Unitless. Basically WBGT but with different weighing of wet bulb and dry temperature

16. Annex 5: Critical appraisal tools

16.1 Revised Cochrane risk-of-bias tool for randomized trials (RoB 2) (86)

Revised Cochrane risk-of-bias tool for randomized trials (RoB 2)
TEMPLATE FOR COMPLETION

Edited by Julian PT Higgins, Jelena Savović, Matthew J Page, Jonathan AC Sterne
on behalf of the RoB2 Development Group

Version of 22 August 2019

The development of the RoB 2 tool was supported by the MRC Network of Hubs for Trials Methodology Research (MR/L004933/2- N61), with the support of the host MRC ConDuCT-II Hub (Collaboration and innovation for Difficult and Complex randomised controlled Trials In Invasive procedures - MR/K025643/1), by MRC research grant MR/M025209/1, and by a grant from The Cochrane Collaboration.



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Study details

Reference	

Study design

X Individually-randomized parallel-group trial

☐ Cluster-randomized parallel-group trial

☐ Individually randomized cross-over (or other matched) trial

For the purposes of this assessment, the interventions being compared are defined as

Experimental:		Comparator:	

Specify which outcome is being assessed for risk of bias	
Specify the numerical result being assessed. In case of multiple alternative analyses being presented, specify the numeric result (e.g. RR = 1.52 (95% CI 0.83 to 2.77) and/or a reference (e.g. to a table, figure or paragraph) that uniquely defines the result being assessed.	

Is the review team's aim for this result…?

☐ to assess the effect of *assignment to intervention* (the 'intention-to-treat' effect)

☐ to assess the effect of *adhering to intervention* (the 'per-protocol' effect)

If the aim is to assess the effect of *adhering to intervention*, select the deviations from intended intervention that should be addressed (at least one must be checked):

☐ occurrence of non-protocol interventions

☐ failures in implementing the intervention that could have affected the outcome

☐ non-adherence to their assigned intervention by trial participants

Which of the following sources were <u>obtained</u> to help inform the risk-of-bias assessment? (tick as many as apply)

☐ Journal article(s) with results of the trial

☐ Trial protocol

☐ Statistical analysis plan (SAP)

☐ Non-commercial trial registry record (e.g. ClinicalTrials.gov record)

☐ Company-owned trial registry record (e.g. GSK Clinical Study Register record)

☐ "Grey literature" (e.g. unpublished thesis)

☐ Conference abstract(s) about the trial

☐ Regulatory document (e.g. Clinical Study Report, Drug Approval Package)

☐ Research ethics application

☐ Grant database summary (e.g. NIH RePORTER or Research Councils UK Gateway to Research)

☐ Personal communication with trialist

☐ Personal communication with the sponsor

Risk of bias assessment

Responses underlined in green are potential markers for low risk of bias, and responses in red are potential markers for a risk of bias. Where questions relate only to sign posts to other questions, no formatting is used.

Domain 1: Risk of bias arising from the randomization process

Signalling questions	Comments	Response options
1.1 Was the allocation sequence random?		Y / PY / PN / N / NI
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?		Y / PY / PN / N / NI
1.3 Did baseline differences between intervention groups suggest a problem with the randomization process?		Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias arising from the randomization process?		NA / Favours experimental / Favours comparator / Towards

		null /Away from null / Unpredictable

Domain 2: Risk of bias due to deviations from the intended interventions (*effect of assignment to intervention*)

Signalling questions	Comments	Response options
2.1. Were participants aware of their assigned intervention during the trial?		Y / PY / PN / N / NI
2.2. Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		Y / PY / PN / N / NI
2.3. If Y/PY/NI to 2.1 or 2.2: Were there deviations from the intended intervention that arose because of the trial context?		NA / Y / PY / PN / N / NI
2.4 If Y/PY to 2.3: Were these deviations likely to have affected the outcome?		NA / Y / PY / PN / N / NI
2.5. If Y/PY/NI to 2.4: Were these deviations from intended intervention balanced between groups?		NA / Y / PY / PN / N / NI
2.6 Was an appropriate analysis used to estimate the effect of assignment to intervention?		Y / PY / PN / N / NI
2.7 If N/PN/NI to 2.6: Was there potential for a substantial impact (on the result) of the failure to analyse participants in the group to which they were randomized?		NA / Y / PY / PN / N / NI

Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias due to deviations from intended interventions?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Domain 2: Risk of bias due to deviations from the intended interventions (*effect of adhering to intervention*)

Signalling questions	**Comments**	**Response options**
2.1. Were participants aware of their assigned intervention during the trial?		Y / PY / PN / N / NI
2.2. Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		Y / PY / PN / N / NI
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?		NA / Y / PY / PN / N / NI
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?		NA / Y / PY / PN / N / NI
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?		NA / Y / PY / PN / N / NI

2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to the intervention?		NA / Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias due to deviations from intended interventions?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Domain 3: Missing outcome data

Signalling questions	Comments	Response options
3.1 Were data for this outcome available for all, or nearly all, participants randomized?		Y / PY / PN / N / NI
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?		NA / Y / PY / PN / N
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?		NA / Y / PY / PN / N / NI

3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?		NA / Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias due to missing outcome data?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Domain 4: Risk of bias in measurement of the outcome

Signalling questions	Comments	Response options
4.1 Was the method of measuring the outcome inappropriate?		Y / PY / PN / N / NI
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?		Y / PY / PN / N / NI
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NA / Y / PY / PN / N / NI

4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		NA / Y / PY / PN / N / NI
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?		NA / Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias in measurement of the outcome?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Domain 5: Risk of bias in selection of the reported result

Signalling questions	**Comments**	**Response options**
5.1 Were the data that produced this result analysed in accordance with a pre-specified analysis plan that was finalized before unblinded outcome data were available for analysis?		Y / PY / PN / N / NI
Is the numerical result being assessed likely to have been selected, on the basis of the results, from...		

5.2. ... multiple eligible outcome measurements (e.g. scales, definitions, time points) within the outcome domain?		Y / PY / PN / N / NI
5.3 ... multiple eligible analyses of the data?		Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias due to selection of the reported result?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Overall risk of bias

Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the overall predicted direction of bias for this outcome?		NA / Favours experimental / Favours comparator / Towards

		null /Away from null / Unpredictable



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16.2 Risk of Bias in Non-randomized Studies – of Interventions (ROBINS-I) assessment tool (87)

The Risk Of Bias In Non-randomized Studies – of Interventions (ROBINS-I) assessment tool
(version for cohort-type studies)

Version 19 September 2016



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ROBINS-I tool (Stage I): At protocol stage

Specify the review question

Participants	
Experimental intervention	
Comparator	
Outcomes	

List the confounding domains relevant to all or most studies

List co-interventions that could be different between intervention groups and that could impact on outcomes

ROBINS-I tool (Stage II): For each study

Specify a target randomized trial specific to the study

Design	Individually randomized / Cluster randomized / Matched (e.g. cross-over)
Participants	
Experimental intervention	
Comparator	

Is your aim for this study…?

☐ to assess the effect of *assignment to* intervention

☐ to assess the effect of *starting and adhering to* intervention

Specify the outcome

Specify which outcome is being assessed for risk of bias (typically from among those earmarked for the Summary of Findings table). Specify whether this is a proposed benefit or harm of intervention.

Specify the numerical result being assessed

In case of multiple alternative analyses being presented, specify the numeric result (e.g. RR = 1.52 (95% CI 0.83 to 2.77) and/or a reference (e.g. to a table, figure or paragraph) that uniquely defines the result being assessed.

Preliminary consideration of confounders

Complete a row for each important confounding domain (i) listed in the review protocol; and (ii) relevant to the setting of this particular study, or which the study authors identified as potentially important.

"Important" confounding domains are those for which, in the context of this study, adjustment is expected to lead to a clinically important change in the estimated effect of the intervention. "Validity" refers to whether the confounding variable or variables fully measure the domain, while "reliability" refers to the precision of the measurement (more measurement error means less reliability).

(i) Confounding domains listed in the review protocol				
Confounding domain	Measured variable(s)	Is there evidence that controlling for this variable was unnecessary?*	Is the confounding domain measured validly and reliably by this variable (or these variables)?	OPTIONAL: Is failure to adjust for this variable (alone) expected to favour the experimental intervention or the comparator?
			Yes / No / No information	Favour experimental / Favour comparator / No information

(ii) Additional confounding domains relevant to the setting of this particular study, or which the study authors identified as important				
Confounding domain	Measured variable(s)	Is there evidence that controlling for this variable was unnecessary?*	Is the confounding domain measured validly and reliably by this variable (or these variables)?	OPTIONAL: Is failure to adjust for this variable (alone) expected to favour the experimental intervention or the comparator?

			Yes / No / No information	Favour experimental / Favour comparator / No information

* In the context of a particular study, variables can be demonstrated not to be confounders and so not included in the analysis: (a) if they are not predictive of the outcome; (b) if they are not predictive of intervention; or (c) because adjustment makes no or minimal difference to the estimated effect of the primary parameter. Note that “no statistically significant association” is not the same as “not predictive”.

Preliminary consideration of co-interventions

Complete a row for each important co-intervention (i) listed in the review protocol; and (ii) relevant to the setting of this particular study, or which the study authors identified as important.

"Important" co-interventions are those for which, in the context of this study, adjustment is expected to lead to a clinically important change in the estimated effect of the intervention.

(i) Co-interventions listed in the review protocol		
Co-intervention	Is there evidence that controlling for this co-intervention was unnecessary (e.g. because it was not administered)?	Is presence of this co-intervention likely to favour outcomes in the experimental intervention or the comparator
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information

(ii) Additional co-interventions relevant to the setting of this particular study, or which the study authors identified as important		
Co-intervention	Is there evidence that controlling for this co-intervention was unnecessary (e.g. because it was not administered)?	Is presence of this co-intervention likely to favour outcomes in the experimental intervention or the comparator
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information

		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information

Risk of bias assessment

Responses underlined in green are potential markers for low risk of bias, and responses in red are potential markers for a risk of bias. Where questions relate only to sign posts to other questions, no formatting is used.

Signalling questions	Description	Response options
Bias due to confounding		
1.1 Is there potential for confounding of the effect of intervention in this study? **If N/PN to 1.1:** the study can be considered to be at low risk of bias due to confounding and no further signalling questions need be considered		Y / PY / PN / N
If Y/PY to 1.1: determine whether there is a need to assess time-varying confounding:		
1.2. Was the analysis based on splitting participants' follow up time according to intervention received? **If N/PN**, answer questions relating to baseline confounding (1.4 to 1.6) **If Y/PY**, go to question 1.3.		NA / Y / PY / PN / N / NI

1.3. Were intervention discontinuations or switches likely to be related to factors that are prognostic for the outcome? **If N/PN**, answer questions relating to baseline confounding (1.4 to 1.6) **If Y/PY**, answer questions relating to both baseline and time-varying confounding (1.7 and 1.8)		NA / Y / PY / PN / N / NI

Questions relating to baseline confounding only		
1.4. Did the authors use an appropriate analysis method that controlled for all the important confounding domains?		NA / Y / PY / PN / N / NI
1.5. **If Y/PY to 1.4**: Were confounding domains that were controlled for measured validly and reliably by the variables available in this study?		NA / Y / PY / PN / N / NI
1.6. Did the authors control for any post-intervention variables that could have been affected by the intervention?		NA / Y / PY / PN / N / NI
Questions relating to baseline and time-varying confounding		

1.7. Did the authors use an appropriate analysis method that controlled for all the important confounding domains and for time-varying confounding?		NA / Y / PY / PN / N / NI
1.8. **If Y/PY to 1.7**: Were confounding domains that were controlled for measured validly and reliably by the variables available in this study?		NA / Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to confounding?		Favours experimental / Favours comparator / Unpredictable

Bias in selection of participants into the study		
2.1. Was selection of participants into the study (or into the analysis) based on participant characteristics observed after the start of intervention? **If N/PN to 2.1:** go to 2.4		Y / PY / PN / N / NI

2.2. **If Y/PY to 2.1**: Were the post-intervention variables that influenced selection likely to be associated with intervention?		NA / Y / PY / PN / N / NI
2.3 **If Y/PY to 2.2**: Were the post-intervention variables that influenced selection likely to be influenced by the outcome or a cause of the outcome?		NA / Y / PY / PN / N / NI
2.4. Do start of follow-up and start of intervention coincide for most participants?		Y / PY / PN / N / NI
2.5. **If Y/PY to 2.2 and 2.3, or N/PN to 2.4**: Were adjustment techniques used that are likely to correct for the presence of selection biases?		NA / Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to selection of participants into the study?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias in classification of interventions		
3.1 Were intervention groups clearly defined?		Y / PY / PN / N / NI
3.2 Was the information used to define intervention groups recorded at the start of the intervention?		Y / PY / PN / N / NI

3.3 Could classification of intervention status have been affected by knowledge of the outcome or risk of the outcome?		Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to classification of interventions?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias due to deviations from intended interventions		
If your aim for this study is to assess the effect of assignment to intervention, answer questions 4.1 and 4.2		
4.1. Were there deviations from the intended intervention beyond what would be expected in usual practice?		Y / PY / PN / N / NI
4.2. **If Y/PY to 4.1**: Were these deviations from intended intervention unbalanced between groups *and* likely to have affected the outcome?		NA / Y / PY / PN / N / NI
If your aim for this study is to assess the effect of starting and adhering to intervention, answer questions 4.3 to 4.6		
4.3. Were important co-interventions balanced across intervention groups?		Y / PY / PN / N / NI
4.4. Was the intervention implemented successfully for most participants?		Y / PY / PN / N / NI

4.5. Did study participants adhere to the assigned intervention regimen?		Y / PY / PN / N / NI
4.6. **If N/PN to 4.3, 4.4 or 4.5**: Was an appropriate analysis used to estimate the effect of starting and adhering to the intervention?		NA / Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to deviations from the intended interventions?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias due to missing data		
5.1 Were outcome data available for all, or nearly all, participants?		Y / PY / PN / N / NI
5.2 Were participants excluded due to missing data on intervention status?		Y / PY / PN / N / NI
5.3 Were participants excluded due to missing data on other variables needed for the analysis?		Y / PY / PN / N / NI
5.4 **If PN/N to 5.1, or Y/PY to 5.2 or 5.3**: Are the proportion of participants and reasons for missing data similar across interventions?		NA / Y / PY / PN / N / NI

5.5 **If PN/N to 5.1, or Y/PY to 5.2 or 5.3**: Is there evidence that results were robust to the presence of missing data?		NA / Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to missing data?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias in measurement of outcomes		
6.1 Could the outcome measure have been influenced by knowledge of the intervention received?		Y / PY / PN / N / NI
6.2 Were outcome assessors aware of the intervention received by study participants?		Y / PY / PN / N / NI
6.3 Were the methods of outcome assessment comparable across intervention groups?		Y / PY / PN / N / NI
6.4 Were any systematic errors in measurement of the outcome related to intervention received?		Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI

Optional: What is the predicted direction of bias due to measurement of outcomes?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias in selection of the reported result		
Is the reported effect estimate likely to be selected, on the basis of the results, from... 7.1. ... multiple outcome *measurements* within the outcome domain?		Y / PY / PN / N / NI
7.2 ... multiple *analyses* of the intervention-outcome relationship?		Y / PY / PN / N / NI
7.3 ... different *subgroups*?		Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to selection of the reported result?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Overall bias

Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the overall predicted direction of bias for this outcome?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

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17. Annex 6: Data Management Plan

18. Annex 7: Protocol Signature Page

Study Title: Individual Participant Data meta-analysis to quantify the impact of high ambient temperatures on maternal and child health in Africa

Study version and date: Protocol v1.0 dated 05 May 2022

I _____ , the investigator, will conduct the study in accordance with the provisions of this protocol and all applicable protocol-related documents, subject to ethical and safety considerations. I agree to comply with ethics committee determinations, all applicable regulations/guidelines and all locally applicable laws.

Investigator of Record Name	Investigator of Record Signature	Date